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SCHEDULING IN ROBOTIC CELLS: CLASSIFICATION, TWO AND THREE MACHINE CELLS

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This paper considers the scheduling of operations in a manufacturing cell that repetitively produces a family of similar parts on two or three machines served by a robot. We provide a classification scheme for scheduling problems in robotic cells. We discuss finding the robot move cycle and the part sequence that jointly minimize the production cycle time, or equivalently maximize the throughput rate. For multiple part-type problems in a two-machine cell, we provide an efficient algorithm that simultaneously optimizes the robot move and part sequencing problems. This algorithm is tested computationally. For a three-machine cell with general data and identical parts, we address an important conjecture about the optimality of repeating one-unit cycles, and show that such a procedure dominates more complicated cycles producing two units. For a three-machine cell producing multiple part-types, we prove that four out of the six potentially optimal robot move cycles for producing one unit allow efficient identification of the optimal part sequence. Several efficiently solvable special cases with practical relevance are identified, since the general problem of minimizing cycle time is intractable. Finally, we discuss ways in which a robotic cell converges to a steady state.

Recent slowed economic growth rates and increasing dependence upon international markets are two factors that have created incentives to improve manufacturing productivity in the world's major trading nations. Attempts to achieve such improvements have increased the level of automation in manufacturing industries, especially in engineering and electronics. Two substantial components of this trend have been cellular manufacturing and robotics. The ideas underlying cellular manufacturing are to identify families of parts that require similar operations, and to produce each family in a separate, highly specialized machine cell. A growing body of evidence suggests that, in a wide variety of industrial settings, material handling within a cell can be accomplished very efficiently by the use of robots. Among the interrelated issues to be considered in using robotic cells are their design, the scheduling of robot moves, and the sequencing of parts to be produced. In this paper, we discuss algorithmic issues related to scheduling in two- and three-machine robotic cells, and scheduling with identical part-types.

The problem of deciding how to form parts and machines into manufacturing cells has been studied by many authors, including McAuley (1972), Burbidge (1975), Rajagopalan and Batra (1975), King (1980), Chan and Milner (1982), Seifoddini and Wolfe (1986), Kusiak (1987), and Askin and Chiu (1990). Much of this work suggests heuristic procedures of a clustering type for forming the cells.

The main focus of the present work, however, is on problems of *scheduling within cells*, rather than of cell formation. Although manufacturing cells can have human operators to perform material handling, the full benefits to productivity from the use of cells are only realized by robot handling. Where this is the case, the manufacturing cell is known as a *robotic cell*. An early implementation at the Fanuc factory in Japan increased cell production from 90 motors per day when manually served, to 300 per day when robot-served (Inaba and Sakakibara 1982). Hartley (1983) points out that a cell system is ideally suited to robot handling, particularly where lot sizes are small. A case study of a three machine robotic cell producing 13 similar components for the defense industry (Hull 1985) showed considerable success in meeting productivity objectives.

There have been several studies of the scheduling of robotic cells. An extensive review of the literature can be found in Sethi et al. (1992). Kondoleon (1979) studies various robot assembly system configurations and their effect on output rate. Baumann et al. (1981) study robot and machine utilization in a robotic cell. Simulation studies of robotic cells are described by Claybourn and Hewit (1982), Badalamenti and Bao (1986), and Noh and Herring (1988). Drezner and Nof (1984) design heuristics for various operations in an assembly cell. Wilhelm and Sarin (1985) consider the parts scheduling problem in robotic

Subject classifications: Production/scheduling: minimizing cycle time in robot-served machine cells. Deterministic, multiple machine sequencing: optimal sequencing of robot moves and of parts.

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cells with various configurations, including parallel identical and nonidentical machines, and flow line manufacturing. However, no optimal scheduling policy is identified. Nof and Drezner (1986) discuss the assembly plan problem, which models layout design in an assembly cell taking into account the necessary operations and movements. A heuristic solution is proposed and tested. The computational complexity of several scheduling problems in assembly cells with one or two robots is discussed by Wilhelm (1987). Rabinowitz et al. (1991) formulate a mixed integer model for a variety of objective functions in a multirobot machining and assembly cell. Unfortunately, the size of their formulation increases very rapidly with the size of the cell. Logendran and Sriskandarajah (1996) study three different robotic cell layouts: the robot-centered cell, the mobile-robot cell, and the in-line robot cell. For given robot move sequences, they identify part sequences for a minimal part set (MPS) that minimize the cycle time, or equivalently maximize the long run production rate. Kise et al. (1991) study the makespan version of the part sequencing problem under a specified move cycle, and describe a polynomial time algorithm.

Sethi et al. (1992) attempt to minimize the cycle time of an MPS. They find the optimal sequence of moves for a two-machine robot-centered cell producing a single part-type. For the three-machine case, they identify the six potentially optimal robot move cycles that produce one unit, and show, under specific assumptions about the data, that two are dominated by the other four. They then provide sufficient conditions under which each of the four non-dominated cycles is optimal. They conjecture that the best robot move cycle for producing one unit provides a cycle time no greater than that given by the best cycle formed from a combination of several one-unit cycles. Finally, they solve the part sequencing problem for a given one-unit robot move cycle in a two machine cell producing multiple part-types. Ioachim and Soumis (1995) study the problem of minimizing cycle time in an m machine robot-centered cell producing a single part-type for a given robot move cycle, using network models. Crama and van de Klundert (1997) describe a polynomial time dynamic programming algorithm for minimizing cycle time over all one-unit cycles in an m machine cell producing a single part-type.

The next three paragraphs summarize results obtained by the authors subsequent to this paper. Hall et al. (1997) consider a three-machine cell producing multiple part-types, and prove that, in two out of the six potentially optimal robot move cycles for producing one unit, the recognition version of the part sequencing problem is unary NP -complete. They show that the general part sequencing problem not restricted to any robot move cycle in a three-machine cell is still intractable.

Kamoun et al. (1994) describe and test simple heuristic procedures for the part sequencing problem in several cell configurations where that problem is intractable. They also study how to design robotic cells for efficient performance by grouping machines into cells, identifying good part

sequences, and providing appropriate size buffers between cells, in a larger manufacturing system.

Sriskandarajah et al. (1994) study a robotic cell containing m machines, for any constant $m \geq 2$. The part sequencing problems associated with a given robot move cycle for producing one unit are classified as follows: (i) sequence independent; (ii) formulated as a traveling salesman problem (TSP); and (iii) unary NP -hard, but not having TSP structure. Some of the problems in class (ii) are a special case which is solvable in polynomial time. As a consequence of this classification, they prove that the part sequencing problems associated with exactly $2m - 2$ of the $m!$ potentially optimal one-unit cycles are polynomially solvable. The remaining cycles have associated part sequencing problems which are unary NP -hard. Also, for three machine robotic cells producing a single part-type, they provide new evidence to support a conjecture discussed in Section 3 of the present paper.

This paper is organized as follows. In Section 1, we briefly describe a robotic cell and its notation. Then we provide a classification scheme for scheduling problems in robotic cells. In all the work that follows, we consider the problem of minimizing cycle time, which is the most natural objective to consider in a repetitive manufacturing environment. In Section 2, an example shows that the best one-unit cycle can be suboptimal in a two-machine cell producing multiple part-types. We therefore provide an efficient algorithm for finding the optimal robot move cycle and the part sequence that jointly minimize the production cycle time. This algorithm is tested computationally. In Section 3 we address an important conjecture, for single part-type production, about the optimality of repeating a simple robot move cycle that produces one unit. For a general cell configuration and general data within the cell, we show through the solution of a mixed integer program that a policy of repeating such one-unit cycles dominates much more complicated cycles producing two units. In Section 4, a number of realistic, easily solvable special cases are identified for part sequencing problems in three machine cells. Since all our results are derived for a robotic cell operating in a steady state, we also study the ways in which a robotic cell converges to such a state. Section 5 contains a conclusion and some suggestions for future work.

1. OVERVIEW AND CLASSIFICATION

We focus on the mobile-robot cell because of its flexibility and widespread application. In a mobile-robot cell, robot travel time may have both linear and rotational components. The mobile-robot cell is a generalization of the in-line robotic cell (where the robot moves linearly), and the robot-centered cell (where robot movement is rotational). Therefore any algorithmic results derived here for the mobile-robot cell will also apply to those other environments. A mobile-robot cell, unlike the other two cells, can be used where the machines cannot be located in a standard configuration around the robot. Such situations arise

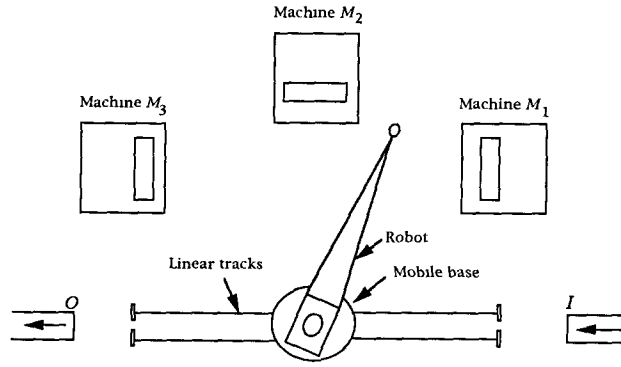


Figure 1. Three-machine mobile-robot cell.

from the size and shape of machines, from the limited reach of the robot arm, and from limited access to floor space. A three machine mobile-robot cell is illustrated in Figure 1. The machines are served by a central robot. Each part being processed passes through the same sequence of locations from the input hopper (I), through machines M_1 , M_2 , and M_3 , and finally into the output hopper (O). Since there are no storage buffers, any part in the cell is always either on one of the machines, or being handled by the robot. Neither the machines nor the robot can be in possession of more than one part at any time. Where a variety of parts is being produced, the processing times on a given machine may vary. All parts may have different processing times on different machines. The processing of any part on a machine is nonpreemptive. The robot travel times, and times to pick up, load, unload, and drop parts, are considered to be independent of the part being moved. Rotational and radial movements of the robot may occur simultaneously, but sometimes additional linear movement is necessary after those movements have been completed. Also, travel times are assumed to be independent of direction. The following notation is used to describe a robotic cell:

- m = the number of machines in the cell.
- $P_1, \dots, P_k$ = the part-types to be produced.
- $r_1, \dots, r_k$ = the minimal ratios of parts to be produced.
- MPS = a minimal part set consisting of r_i parts of type P_i , $i = 1, \dots, k$.
- $n = r_1 + \dots + r_k$ = the total number of parts to be produced in the MPS. (Note that since the part set is minimal, the numbers of parts of different types are equal to the ratios of parts.)
- $a_i, b_i, c_i, \dots$ = the processing times of part i on machines $M_1, M_2, M_3, \dots$, respectively.
- δ_0 = the time taken by additional linear robot movement when traveling between two consecutive machines M_i and M_{i+1} , $1 \leq i \leq m - 1$, after

rotational movement has been completed. In a robot-centered cell, $\delta_0 = 0$.

δ_i = the time taken by a rotational robot movement when traveling between two consecutive machines M_{i-1} and M_i , $1 \leq i \leq m + 1$, where we take I to be machine M_0 , and O to be machine M_{m+1} .

γ = the time saved by not stopping at an intermediate point during a robot movement between two nonconsecutive machines.

ϵ_1 = the time taken by the robot to pick up a part at I .

ϵ_{2i} = the time taken by the robot to load a part onto machine M_i .

ϵ_{2i+1} = the time taken by the robot to unload a part from machine M_i .

ϵ_{2m+2} = the time taken by the robot to drop a part at O .

$(\chi_1, \dots, \chi_m, M_h^j)$ = the current state of the system, where $\chi_i = \phi$ (respectively, Ω) if machine M_i is free (resp., occupied by a part), and the robot has just loaded (respectively, unloaded) machine M_h if $j = -$ (resp., $+$).

The standard classification scheme for scheduling problems (Graham et al. 1979) can be denoted $\psi_1|\psi_2|\psi_3$, where ψ_1 indicates the scheduling environment, ψ_2 describes the job characteristics or restrictive requirements, and ψ_3 defines the objective function to be minimized. Here we extend this scheme to provide for the classification of scheduling problems arising in robotic cells, as follows: Under ψ_1 , we may have

- MRCm = a mobile-robot cell with m machines.
- RCCm = a robot-centered cell with m machines.
- IRCM = an in-line robot cell with m machines.

Under ψ_2 , we may have

- k = the number of part-types.
- r - unit(s) = the problem is being solved over robot move cycles that produce r units.
- S_i = robot move cycle S_i alone is used.
- $\delta_i = \delta$ = the travel time between any pair of consecutive machines is equal.
- $\epsilon_i = \epsilon$ = the load and unload times at all machines are equal.

Under ψ_3 , we may have

- C_i = the average steady state cycle time for the repetitive manufacture of an MPS.
- $C_{\max}$ = the makespan for the manufacture of a given set of jobs.

Thus, for example, $\text{RCC3}|k = 1, S_1|C_i$ denotes the minimization of cycle time for a single part (since $k = 1$, an MPS

consists of one part) in a three-machine robot-centered cell producing one part-type, restricted to robot move cycle S_1 .

2. TWO-MACHINE CELL, MULTIPLE PART-TYPES

We begin our analysis of the two-machine robotic cell problem, $MRC2|k \geq 2|C_r$, by showing that the optimal solution is not generally given by a one-unit cycle, in contrast with the discussion in Section 3 for single part-type problems. The cycle time minimization problem in the multiple part-types case then becomes one of deciding which parts to process under the two possible robot move cycles, S_1 and S_2 , described by Sethi et al. (1992) and below, and how to switch from one cycle to another, while simultaneously choosing the optimal part sequence. An $O(n^4)$ time algorithm which solves this problem optimally is provided.

Let σ denote the sequence of parts of an MPS that have to be processed repetitively, and $T_{j\sigma(i)\sigma(i+1)}^h$ the time between the loading of part $P_{\sigma(i)}$ on machine M_h , and the loading of part $P_{\sigma(i+1)}$ on M_h , using robot move cycle S_j , where $j \in \{1, 2\}$ and $h \in \{1, 2\}$. Note that $P_{\sigma(i)}$ is the i^{th} part in the sequence σ to be produced. We use w_j^i to denote the waiting time of the robot before unloading part $P_{\sigma(i)}$ at machine M_j . We now derive cycle time expressions for S_1 and S_2 .

Starting from the initial state $E = (\phi, \Omega, M_2^-)$, where the robot has just loaded part $P_{\sigma(i)}$ on M_2 , robot move cycle S_1 includes the following activities, with times shown in parentheses: wait for $P_{\sigma(i)}$ at M_2 : $(b_{\sigma(i)})$, unload: (ϵ_5) , move to O : (δ_3) , drop $P_{\sigma(i)}$: (ϵ_6) , move to I : $(\delta_0 + \delta_1 + \delta_2 + \delta_3 - 2\gamma)$, pick up $P_{\sigma(i+1)}$: (ϵ_1) , move to M_1 : (δ_1) , load: (ϵ_2) , wait for $P_{\sigma(i+1)}$: $(a_{\sigma(i+1)})$, unload: (ϵ_3) , move to M_2 : $(\delta_0 + \delta_2)$, and load: (ϵ_4) . Thus $T_{1\sigma(i)\sigma(i+1)}^2 = b_{\sigma(i)} + \epsilon_5 + \delta_3 + \epsilon_6 + (\delta_0 + \delta_1 + \delta_2 + \delta_3 - 2\gamma) + \epsilon_1 + \delta_1 + \epsilon_2 + a_{\sigma(i+1)} + \epsilon_3 + (\delta_0 + \delta_2) + \epsilon_4$.

Starting from the initial state $E = (\phi, \Omega, M_2^-)$, where the robot has just loaded part $P_{\sigma(i)}$ on M_2 , robot move sequence S_2 includes the following activities: move to I : $(\delta_0 + \delta_1 + \delta_2 - \gamma)$, pick up part $P_{\sigma(i+1)}$: (ϵ_1) , move to M_1 : (δ_1) , load: (ϵ_2) , move to M_2 : $(\delta_0 + \delta_2)$, if necessary wait for $P_{\sigma(i)}$: (w_2^i) , unload: (ϵ_5) , move to O : (δ_3) , drop $P_{\sigma(i)}$: (ϵ_6) , move to M_1 : $(\delta_0 + \delta_2 + \delta_3 - \gamma)$, if necessary wait for $P_{\sigma(i+1)}$: (w_1^{i+1}) , unload: (ϵ_3) , move to M_2 : $(\delta_0 + \delta_2)$, and load: (ϵ_4) . Therefore $T_{2\sigma(i)\sigma(i+1)}^2 = (\delta_0 + \delta_1 + \delta_2 - \gamma) + \epsilon_1 + \delta_1 + \epsilon_2 + (\delta_0 + \delta_2) + w_2^i + \epsilon_5 + \delta_3 + \epsilon_6 + (\delta_0 + \delta_2 + \delta_3 - \gamma) + w_1^{i+1} + \epsilon_3 + (\delta_0 + \delta_2) + \epsilon_4$, where $w_1^{i+1} = \max\{0, a_{\sigma(i+1)} - w_2^i - \epsilon_5 - \epsilon_6 - 2\delta_0 - 2\delta_2 - 2\delta_3 + \gamma\}$, and $w_2^i = \max\{0, b_{\sigma(i)} - \epsilon_1 - \epsilon_2 - 2\delta_0 - 2\delta_1 - 2\delta_2 + \gamma\}$. By combining the expressions for w_1^{i+1} and w_2^i , we obtain the following result.

Lemma 1. *The cycle times for one unit in $MRC2|k \geq 2|C_r$ are given by*

$$T_{1\sigma(i)\sigma(i+1)}^2 = \rho + b'_{\sigma(i)} + a'_{\sigma(i+1)},$$

$$T_{2\sigma(i)\sigma(i+1)}^2 = \rho + \max\{\mu, b'_{\sigma(i)}, a'_{\sigma(i+1)}\},$$

Table I
Example Instance in Lemma 2

i	1	2	3	4	5
a_i	7	397	7	397	7
b_i	397	17	397	7	7

where $a'_i = a_i + \epsilon_1 + \epsilon_2 + 2\delta_1 - \gamma$, $b'_i = b_i + \epsilon_5 + \epsilon_6 + 2\delta_3 - \gamma$, $i = 1, \dots, n$, $\rho = \epsilon_3 + \epsilon_4 + 2\delta_0 + 2\delta_2$, and $\mu = 2 \sum_{i=0}^3 \delta_i + \epsilon_1 + \epsilon_2 + \epsilon_5 + \epsilon_6 - 2\gamma$.

Proof. Follows from the above analysis. $\square$

In both cycles S_1 and S_2 , a pair of parts $P_{\sigma(i)}$, $P_{\sigma(i+1)}$ is involved. The reader should also note that the state $E = (\phi, \Omega, M_2^-)$, in which the robot has just finished loading a part on M_2 , is the only state common to both cycles S_1 and S_2 . Hence, this is the only state in which switching between S_1 and S_2 can be achieved without wasteful robot moves. In $E = (\phi, \Omega, M_2^-)$, the robot has two choices:

- wait and unload M_2 (as in cycle S_1);
- move to I (as in cycle S_2).

Because of the possibility of switching between cycles at $E = (\phi, \Omega, M_2^-)$, a part $P_{\sigma(i)}$ may be processed using both cycles. The following possibilities exist:

- During the processing of $P_{\sigma(i)}$ on both machine M_1 and machine M_2 , the robot uses cycle S_i , $i \in \{1, 2\}$. Here we say that $P_{\sigma(i)}$ is processed using cycle S_i .
- The robot uses cycle S_1 during the processing of $P_{\sigma(i)}$ on M_1 , and cycle S_2 during processing on M_2 . We say that $P_{\sigma(i)}$ is processed using cycle $S_{1,2}$.
- The robot uses cycle S_2 during the processing of $P_{\sigma(i)}$ on M_1 , and cycle S_1 during processing on M_2 . We say that $P_{\sigma(i)}$ is processed using cycle $S_{2,1}$.

Lemma 2. *Repetition of the best one-unit robot move cycle is not generally optimal for problem $MRC2|k \geq 2|C_r$.*

Proof. Consider the following instance of $MRC2|k \geq 2|C_r$: $n = 5$; $k = 4$; $\delta_0 = 7$, $\delta_1 = 1$, $\delta_2 = 10$, $\delta_3 = 1$; $\epsilon_i = 1$, $i = 1, \dots, 6$; $\gamma = 1$; the processing times are shown in Table I. From Lemma 1, the time to produce the parts in an MPS in the sequence σ under robot move cycle S_1 is $T_{1(\sigma)} = \sum_{i=1}^n T_{1\sigma(i)\sigma(i+1)}^2 = n\rho + \sum_{i=1}^n (a'_i + b'_i) = 5[1 + 1 + 14 + 20] + [10 + 400 + 10 + 400 + 10] + [400 + 20 + 400 + 10 + 10] = 1850$ for any sequence σ .

The part sequencing problem under robot move cycle S_2 can be transformed into the problem $F_2|no - wait|C_r$ having the form: minimize $Z + \sum_{i=1}^n \max\{e_{\sigma(i+1)}, f_{\sigma(i)}\}$, where Z is a constant, and $e_{\sigma(i)} = a'_{\sigma(i)}$, $f_{\sigma(i)} = \max\{\mu, b'_{\sigma(i)}\}$ are the processing times for part $P_{\sigma(i)}$ on the first machine (M'_1) and the second machine (M'_2), respectively. In this problem, the operations of any job are performed continuously without waiting on or between machines. Recall the rule from Gilmore and Gomory (1964), that a job sequence is optimal if the shortest processing time on M'_1 is

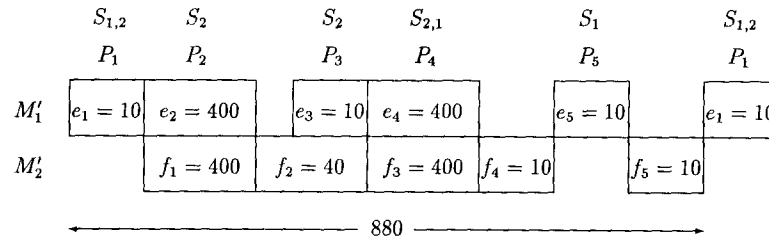


Figure 2. Combining robot move cycles in Lemma 2.

concurrent with that on M'_2 , similarly for the second shortest processing times on the two machines, and so on. For example, processing times e_2 and f_1 are concurrent in Figure 2, as are e_3 and f_2 . Using cycle S_2 , the part sequence $\{1, 2, 3, 4, 5\}$ satisfies this rule, and is therefore optimal. From Lemma 1, the time to produce the parts in an MPS in the sequence σ under robot move cycle S_2 is $T_{2(\sigma)} = \sum_{i=1}^n T_{2\sigma(i)\sigma(i+1)}^2 = n\rho + \sum_{i=1}^n \max\{\mu, b'_{\sigma(i)}, a'_{\sigma(i+1)}\} = 180 + 400 + 40 + 400 + 40 + 40 = 1100$.

The use of a combination of different cycles in the two machine flowshop, using the same part sequence, is illustrated in Figure 2. The part sequence is optimal for this combination, where part P_1 is processed under cycle $S_{1,2}$, parts P_2 and P_3 under cycle S_2 , part P_4 under cycle $S_{2,1}$, and part P_5 under cycle S_1 . The length of each operation is shown by its $e_{\sigma(i)}$ or $f_{\sigma(i)}$ value, and the contribution of unavoidable robot travel time (i.e., $n\rho = 5(36) = 180$) is ignored. Note that $f_{\sigma(i)} = \max\{\mu, b'_{\sigma(i)}\}$ if part $P_{\sigma(i)}$ is processed by cycle S_2 or cycle $S_{1,2}$, otherwise $f_{\sigma(i)} = b'_{\sigma(i)}$. Adding in the travel time, we then have a total cycle time of $880 + 180 = 1060 < \min\{1850, 1100\}$. Intuitively, we have taken advantage of the fact that some parts are suitable for processing under S_1 , and others under S_2 . This completes the proof of Lemma 2. $\square$

Remark 1. The procedure used by Sethi et al. (1992) for this problem involves fixing the robot move cycle to S_1 and solving for the optimal part sequence, and then repeating that process for S_2 , before comparing the two cycle times thus generated. Lemma 2 shows that we can improve the cycle time by using both sequences S_1 and S_2 in the repetitive production of an MPS. A further improvement can be achieved by using cycle $S_{2,1}$ for part P_2 and cycle $S_{1,2}$ for part P_3 . The cycle time then becomes $870 + 180 = 1050$. This demonstrates that an optimal solution may need more than one transition from S_1 to S_2 , and vice versa, in $\text{MRC2}[k \geq 2|C_r]$. Note that Kise et al. (1991) study the makespan version of the part sequencing problem under an Automated Guided Vehicle (AGV) cycle equivalent to S_2 . Their problem is equivalent to $\text{MRC2}[k \geq 2, S_2|C_{\max}]$, and they describe an $O(n^3)$ time algorithm. See Ganesharajah et al. (1997) for more details.

We describe below an algorithm, **MinCycle**, that jointly optimizes the robot move cycle and part sequence for problem $\text{MRC2}[k \geq 2|C_r]$. The idea behind the algorithm is as follows. **MinCycle** begins by finding an arbitrary part

sequence using robot move cycle S_1 (Step 1), and an optimal part sequence (from the algorithm of Gilmore and Gomory 1964) using robot move cycle S_2 (Step 4), and comparing their cycle times (at Step 2 of Subroutine **GG**). Between these two extreme solutions are many which use a mixture of robot move cycles, and these are investigated in Step 5. In this step, the algorithm enumerates $O(n^3)$ ways of setting q of the $a'_{\theta(i)}$ values and q of the b'_i values to a large number, H , where $q = 1, \dots, t$, $t = \min\{|\{i|a'_i < \mu\}|, |\{i|b'_i < \mu\}|\}$, and $H = n\rho + \sum_{i=1}^n (a'_i + b'_i) + 1$. In Step 5, the $(q-1)$ smallest $a'_{\theta(i)}$ s and the $(q-1)$ smallest b'_i s are set to H . It remains to set one $a'_{\theta(i)}$ value and one b'_i value to H , and all possible such assignments are examined. This requires $O(n^2)$ time for fixed q . Each assignment of the qH s corresponds to a partition of the MPS into four subsets B_1, B_2, B_{12} and B_{21} consisting of parts produced using robot move cycles $S_1, S_2, S_{1,2}$ and $S_{2,1}$, respectively. If a part P_i has both a'_i and b'_i set to H , then that part belongs to B_1 . If only a'_i (respectively, b'_i) is set to H , then that part belongs to B_{12} , (resp., B_{21}). If neither a'_i nor b'_i is set to H , then that part belongs to B_2 . For each assignment of the qH s, Subroutine **GG** is called to solve the equivalent $F_2|no\text{-}wait|C_r$ problem. Following an optimality test (Step 2 of **GG**), the smallest cycle time C^* , and corresponding schedule σ^* and partition B_1, B_2, B_{12} , and B_{21} obtained from Steps 1, 4, and 5 are recorded as an optimal solution.

Note that since H is large, Subroutine **GG** provides a schedule in which each a'_i set to H is processed on M'_1 concurrently with a b'_j set to H on M'_2 . This pairing in **GG** means that the processing times corresponding to these two H values, e_i and f_j , can be represented in the schedule as two consecutive blocks similar to f_4 and e_5 in Figure 2. Therefore, in any schedule, the sequence S_1 or $S_{2,1}$ is followed by either S_1 or $S_{1,2}$. Similarly, the sequence S_2 or $S_{1,2}$ is always followed by either S_2 or $S_{2,1}$. Step 2 of **GG** calculates the actual cycle time of each partition defined by the assignment of the H values in Step 5 of **MinCycle**.

The Gilmore and Gomory algorithm solves a special case of the traveling salesman problem with the cost structure $c_{ij} = \max\{e_j, f_i\}$. It also solves the two machine no-wait flowshop cycle time problem, $F_2|no\text{-}wait|C_r$, in $O(n \log n)$ time, since that problem has the same cost structure (Reddi and Ramamoorthy 1972, Lawler et al. 1985). In the latter problem a given set of jobs is produced repetitively, once each cycle. The intuition behind the algorithm is that,

ideally, the shortest processing time on machine M'_1 would be concurrent with that on machine M'_2 , similarly for the second shortest processing times on the two machines, and so on. If this is not possible, a dual improvement step moves the current schedule towards feasibility at minimum cost. The detailed steps of the Gilmore and Gomory algorithm, and their application to an example problem, are described by Hall and Sris-kandarajah (1996). Our algorithm for $MRC2|k \geq 2|C_t|$ now follows.

Algorithm MinCycle

Step 0: Given $a_i, b_i, 1 \leq i \leq n, \delta_0, \dots, \delta_3$ and γ , and $\epsilon_1, \dots, \epsilon_6$
 $B_1 = \{P_1, \dots, P_n\}, B_2 = B_{12} = B_{21} = \emptyset$
 $\rho = \epsilon_3 + \epsilon_4 + 2\delta_0 + 2\delta_2$
 $\mu = 2 \sum_{i=0}^3 \delta_i + \epsilon_1 + \epsilon_2 + \epsilon_5 + \epsilon_6 - 2\gamma$
 Step 1: $a'_i = a_i + \epsilon_1 + \epsilon_2 + 2\delta_1 - \gamma, i = 1, \dots, n$
 $b'_i = b_i + \epsilon_5 + \epsilon_6 + 2\delta_3 - \gamma, i = 1, \dots, n$
 $\sigma^* = \{1, \dots, n\}$
 $C^* = n\rho + \sum_{i=1}^n (a'_i + b'_i)$
 $H = C^* + 1$
 $Y = 0, q = 0$
 Step 2: Number the parts so that $b'_i \leq \dots \leq b'_n$
 Find an ordering θ of the parts such that $a'_{\theta(1)} \leq \dots \leq a'_{\theta(n)}$
 Step 3: $t = \min\{|\{i|a'_i < \mu\}|, |\{i|b'_i < \mu\}|\}$
 Step 4: Call **GG**($a'_i, b'_i, 1 \leq i \leq n$)
 Step 5: For $q = 1, \dots, t$ do
 $Y = \sum_{i=1}^{q-1} (a'_{\theta(i)} + b'_i)$
 $a'_{\theta(i)} = H, b'_i = H, i = 1, \dots, q - 1$
 For $h, j = q, \dots, n$ do
 $W_a = a'_{\theta(h)}, W_b = b'_j$
 $a'_{\theta(h)} = H, b'_j = H$
 $Y = Y + W_a + W_b$
 Call **GG**($a'_i, b'_i, 1 \leq i \leq n$)
 $a'_{\theta(h)} = W_a, b'_j = W_b$
 $Y = Y - W_a - W_b$
 End
 End
 Stop.

Subroutine GG ($a'_i, b'_i, 1 \leq i \leq n$)

Step 1: $e_i = a'_i, f_i = \max\{\mu, b'_i\}, i = 1, \dots, n$
 Apply the algorithm of Gilmore and Gomory (1964) to the instance $\{e_i, f_i\}, i = 1, \dots, n$
 Let σ denote the optimal part sequence, and C its value
 Step 2: If $C + n\rho - qH + Y < C^*$, then
 $C^* = C + n\rho - qH + Y, \sigma^* = \sigma$
 $B_1 = \{P_i|a'_i = H\} \cap \{P_i|b'_i = H\}$
 $B_{12} = \{P_i|a'_i = H\} \setminus \{P_i|b'_i = H\}$
 $B_{21} = \{P_i|b'_i = H\} \setminus \{P_i|a'_i = H\}$
 $B_2 = \{P_1, \dots, P_n\} \setminus (B_1 \cup B_{12} \cup B_{21})$
 End if
 Return.

In order to prove the optimality of **MinCycle**, we need several preliminary results.

Lemma 3. In any feasible solution for $MRC2|k \geq 2|C_t|, |B_{12}| = |B_{21}|$.

Proof. In general, any feasible solution is a cyclic production of an MPS in a sequence in which each part is produced using one of the following robot move cycles: $S_1, S_2, S_{1,2}$, and $S_{2,1}$. Note that cycle $S_{1,2}$ must be preceded by S_1 or $S_{2,1}$, and followed by S_2 or $S_{2,1}$. Since the schedule is cyclic, it follows that the number of transitions from S_1 into S_2 (using $S_{1,2}$) must equal the number of transitions from S_2 into S_1 (using $S_{2,1}$). $\square$

Lemma 4. Algorithm **MinCycle** evaluates all solutions in which $|B_1 \cup B_{12}| = |B_1 \cup B_{21}| = q$, where $q = 0$ if the solution is found in Step 4, and $1 \leq q \leq t$ if the solution is found in Step 5, and where the $\max\{0, q - 1\}$ smallest a'_i (respectively, b'_i) values are associated with parts in $B_1 \cup B_{12}$ (resp., $B_1 \cup B_{21}$).

Proof. This follows from Steps 4 and 5 of **MinCycle**, and Step 2 of **GG**. $\square$

Lemma 5. For any real numbers $v, w, x \geq 0$, with $w \geq v$, $\max\{x, v\} + w \geq \max\{x, w\} + v$.

Proof. $\max\{x, w\} = \max\{x, v + (w - v)\} \leq \max\{x, v\} + (w - v)$. $\square$

These lemmas will be useful in the proof of the main result, which now follows.

Theorem 1. Algorithm **MinCycle** finds an optimal solution to $\text{MRC2}[k \geq 2|C_i]$ in $O(n^4)$ time.

Proof. The first part of the proof shows that, for given subsets B_1, B_2, B_{12} and B_{21} , **MinCycle** finds an optimal part sequence, σ , whose cycle time is denoted by T_σ . As we observe above, during this sequence the robot may switch between robot move cycles S_1 and S_2 . We write the cycle time as $T_\sigma = n\rho + \sum_{i=1}^n d_{\sigma(i)\sigma(i+1)}$, where we define $d_{\sigma(n)\sigma(n+1)} = d_{\sigma(n)\sigma(1)}$ and

$$T_\sigma = n\rho + \sum_{i=1}^n V_{\sigma(i)\sigma(i+1)} + \sum_{i=1}^n W_{\sigma(i)\sigma(i+1)}, \quad \text{where}$$

$$d_{\sigma(i)\sigma(i+1)} = V_{\sigma(i)\sigma(i+1)} \\ = \begin{cases} b'_{\sigma(i)} + a'_{\sigma(i+1)} & \text{if } S_1 \text{ is used for the pair} \\ & P_{\sigma(i)}, P_{\sigma(i+1)}, \\ 0 & \text{otherwise.} \end{cases}$$

$$d_{\sigma(i)\sigma(i+1)} = W_{\sigma(i)\sigma(i+1)} \\ = \begin{cases} \max\{\mu, b'_{\sigma(i)}, a'_{\sigma(i+1)}\} & \text{if } S_2 \text{ is used for the pair} \\ & P_{\sigma(i)}, P_{\sigma(i+1)}, \\ 0 & \text{otherwise.} \end{cases}$$

For these definitions, S_1 is used for the pair $P_{\sigma(i)}, P_{\sigma(i+1)}$ if and only if $P_{\sigma(i)} \in B_1 \cup B_{21}$ and $P_{\sigma(i+1)} \in B_1 \cup B_{12}$. A similar characterization holds for S_2 . Since $n\rho$ is a constant, the above problem is equivalent to minimizing

$$\bar{T}_\sigma = \sum_{i=1}^n V_{\sigma(i)\sigma(i+1)} + \sum_{i=1}^n W_{\sigma(i)\sigma(i+1)}.$$

Define $Y = \sum_{P_i \in B_1 \cup B_{12}} a'_i + \sum_{P_i \in B_1 \cup B_{21}} b'_i$. Now letting $b'_{\sigma(i)} = a'_{\sigma(i+1)} = H$ whenever $V_{\sigma(i)\sigma(i+1)} = b'_{\sigma(i)} + a'_{\sigma(i+1)}$ gives

$$\hat{T}_\sigma = \sum_{i=1}^n \max\{f_{\sigma(i)}, e_{\sigma(i+1)}\},$$

where $f_{\sigma(j)} = \max\{\mu, b'_{\sigma(j)}\}$ and $e_{\sigma(j)} = a'_{\sigma(j)}$, $j = 1, \dots, n$. It remains to show that the sequence which minimizes $\hat{T}_\sigma$ also minimizes T_σ . To see this, note that since H has a large value, each a'_i set to H on M'_i is concurrent with a b'_i set to H on M'_i in any optimal solution. Thus $qH \leq \hat{T}_\sigma <$

$(q + 1)H$. Step 2 of **GG** ignores sequences σ for which $\hat{T}_\sigma \geq (q + 1)H$, since the optimal solution can be found elsewhere. Thus, in σ^* , no part in B_{12} can immediately precede a part in B_1 or B_{12} , since this would imply that $\hat{T}_\sigma \geq (q + 1)H$. Similarly, no part in B_{21} can immediately precede a part in B_2 or B_{21} in σ^* . Therefore, if $\hat{T}_\sigma < (q + 1)H$, then

$$\hat{T}_\sigma = T_\sigma - n\rho + qH - Y,$$

hence the two problems are equivalent under that condition.

The second part of the proof shows that the partitions of the parts into subsets B_1, B_2, B_{12} , and B_{21} considered by **MinCycle** contain an optimal schedule. In order to characterize such partitions, let (π^*, σ^*) represent an optimal solution, consisting of a partition π^* of the parts into B_1, B_2, B_{12} , and B_{21} , and a part sequence σ^* . We make use of two claims.

Claim 1. If $a'_{\sigma^*(i+1)} + b'_{\sigma^*(i)} < \mu$, then $d_{\sigma^*(i)\sigma^*(i+1)} = V_{\sigma^*(i)\sigma^*(i+1)}$, and if $a'_{\sigma^*(i+1)} + b'_{\sigma^*(i)} \geq \mu$, then $d_{\sigma^*(i)\sigma^*(i+1)} = W_{\sigma^*(i)\sigma^*(i+1)}$.

Proof of Claim 1. Assume that there exist parts $P_{\sigma^*(i)}$ and $P_{\sigma^*(i+1)}$ such that $d_{\sigma^*(i)\sigma^*(i+1)} = W_{\sigma^*(i)\sigma^*(i+1)}$ and $a'_{\sigma^*(i+1)} + b'_{\sigma^*(i)} < \mu = \max\{\mu, b'_{\sigma^*(i)}, a'_{\sigma^*(i+1)}\} = W_{\sigma^*(i)\sigma^*(i+1)}$. Now if we form a new solution (π', σ') by setting $d_{\sigma^*(i)\sigma^*(i+1)} = V_{\sigma^*(i)\sigma^*(i+1)} = b'_{\sigma^*(i)} + a'_{\sigma^*(i+1)}$, we find a new cycle time, $T_{\pi'\sigma'}$, satisfying $T_{\pi'\sigma'} - T_{\pi^*\sigma^*} = b'_{\sigma^*(i)} + a'_{\sigma^*(i+1)} - \mu \Rightarrow T_{\pi'\sigma'} < T_{\pi^*\sigma^*}$, contradicting the optimality of (π^*, σ^*) . Note that the change from π^* to π' alters the robot move cycles for parts $P_{\sigma^*(i)}$ and $P_{\sigma^*(i+1)}$. The proof of the second part is similar.

Claim 2. If there exists an optimal solution (π^*, σ^*) with $|B_1 \cup B_{12}| = |B_1 \cup B_{21}| = q$, then there exists an optimal solution satisfying not only that condition, but also the following two:

(a) The $(q - 1)$ smallest a'_i values are associated with parts in B_1 or B_{12} , thus $d_{\sigma^*(i-1)\sigma^*(i)} = V_{\sigma^*(i-1)\sigma^*(i)} = b'_{\sigma^*(i-1)} + a'_{\sigma^*(i)}$

(b) The $(q - 1)$ smallest b'_i values are associated with parts in B_1 or B_{21} , thus $d_{\sigma^*(i)\sigma^*(i+1)} = V_{\sigma^*(i)\sigma^*(i+1)} = b'_{\sigma^*(i)} + a'_{\sigma^*(i+1)}$.

Proof of Claim 2. For condition (a), assume that in (π^*, σ^*) , part $P_{\sigma^*(j)} \in B_2 \cup B_{21}$, where $a'_{\sigma^*(j)}$ is one of the $(q - 1)$ smallest a'_i values. It follows that there exist parts $P_{\sigma^*(u)}, P_{\sigma^*(v)} \in B_1 \cup B_{12}$ such that $a'_{\sigma^*(u)}, a'_{\sigma^*(v)} \geq a'_{\sigma^*(j)}$. Without loss of generality, let $\sigma^* = \{P_{\sigma^*(j)}, \sigma^*_{ju}, P_{\sigma^*(u)}, \sigma^*_{uv}, P_{\sigma^*(v)}, \sigma^*_{vj}\}$, where $\sigma^*_{ju} = \{P_{\sigma^*(j+1)}, P_{\sigma^*(j+2)}, \dots, P_{\sigma^*(u-2)}, P_{\sigma^*(u-1)}\}$ denote a partial sequence of jobs in between the j th and u th positions of σ^* , as in Figure 3. Since parts $P_{\sigma^*(u)}, P_{\sigma^*(v)} \in B_1 \cup B_{12}$, it follows that parts $P_{\sigma^*(u-1)}, P_{\sigma^*(v-1)} \in B_1 \cup B_{21}$. Similarly, since part $P_{\sigma^*(j)} \in B_2 \cup B_{21}$, it follows that part $P_{\sigma^*(j-1)} \in B_2 \cup B_{12}$. Thus we can write $d_{\sigma^*(j-1)\sigma^*(j)} = \max\{\mu, b'_{\sigma^*(j-1)}, a'_{\sigma^*(j)}\}$, $d_{\sigma^*(u-1)\sigma^*(u)} = b'_{\sigma^*(u-1)} + a'_{\sigma^*(u)}$, and $d_{\sigma^*(v-1)\sigma^*(v)} = b'_{\sigma^*(v-1)} + a'_{\sigma^*(v)}$. Let (π', σ') be another solution with $\sigma' = \{P_{\sigma^*(u)}, \sigma^*_{uv},$

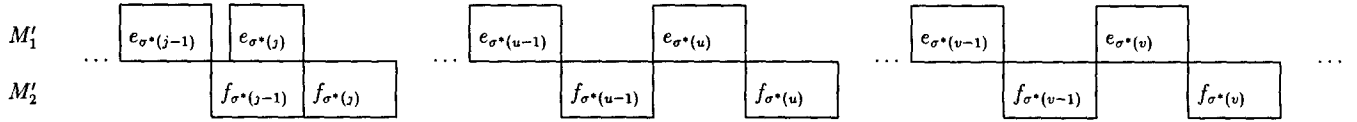


Figure 3. Partition π^* and sequence $\sigma^* = \{P_{\sigma^*(j)}, \sigma_{ju}^*, P_{\sigma^*(u)}, \sigma_{uv}^*, P_{\sigma^*(v)}, \sigma_{vj}^*\}$ in condition (a) of Theorem 1.

$P_{\sigma^*(j)}, \sigma_{ju}^*, P_{\sigma^*(v)}, \sigma_{vj}^*\}$, as in Figure 4. In σ' , all parts belong to the same subsets as in σ^* , except $P_{\sigma^*(j)} \in B_1 \cup B_{12}$ and $P_{\sigma^*(u)} \in B_2 \cup B_{21}$. Thus, in σ' , we have $d_{\sigma^*(j-1)\sigma^*(u)} = \max\{\mu, b'_{\sigma^*(j-1)}, a'_{\sigma^*(u)}\}$, $d_{\sigma^*(v-1)\sigma^*(j)} = b'_{\sigma^*(v-1)} + a'_{\sigma^*(j)}$, and $d_{\sigma^*(u-1)\sigma^*(v)} = b'_{\sigma^*(u-1)} + a'_{\sigma^*(v)}$. Comparing the cycle times of the two solutions, we have $T_{\pi'\sigma'} - T_{\pi^*\sigma^*} = \max\{\mu, a'_{\sigma^*(u)}, b'_{\sigma^*(j-1)}\} + b'_{\sigma^*(v-1)} + a'_{\sigma^*(j)} + b'_{\sigma^*(u-1)} + a'_{\sigma^*(v)} - \max\{\mu, a'_{\sigma^*(j)}, b'_{\sigma^*(j-1)}\} - b'_{\sigma^*(u-1)} - a'_{\sigma^*(u)} - b'_{\sigma^*(v-1)} - a'_{\sigma^*(v)} = \max\{\mu, a'_{\sigma^*(u)}, b'_{\sigma^*(j-1)}\} + a'_{\sigma^*(j)} - \max\{\mu, a'_{\sigma^*(j)}, b'_{\sigma^*(j-1)}\} - a'_{\sigma^*(u)} \leq 0$, from Lemma 5. It follows that, without increasing the cycle time, we can repeat such changes until condition (a) is satisfied.

The proof for condition (b) is similar due to the following symmetrical property of the schedule. If we interchange the roles of M'_1 and M'_2 and consider any schedule in the reverse direction, we find an equivalent schedule where b'_i (resp., a'_i) is the processing time on M'_1 (resp., M'_2).

We are now able to prove the theorem. Steps 4 and 5 of **MinCycle** consider different values of q . If an optimal solution exists at a given value of q , Lemma 3 and Claim 2 define conditions present in at least one optimal solution. Lemma 4 proves that all possible partitions of the parts using those conditions are examined in Step 4 for $q = 0$, and in Step 5 for $1 \leq q \leq t$. It follows from Claim 1 and the definition of t in Step 3 that setting a value H to any processing time of a part with index greater than t will not improve the cycle time. Thus there exists an optimal solution with $0 \leq q \leq t$. Finally, given any partition, the first part of the proof establishes that Subroutine **GG** finds an optimal part sequence.

The dominant step in **MinCycle** is Step 5, where the indices h, j and q are all $O(n)$. The number of calls of **GG** is therefore $O(n^3)$. Lawler et al. (1985) and Matsuo (1990) point out that Step 1 of **GG** can be implemented in $O(n \log n)$ time. However, the parts have already been ordered in Step 4 of **MinCycle**. Consequently, in Step 5, we only need to order the jobs whose processing times are set to a value of H . Since all such jobs have to appear in an arbitrary subsequence at the end of the ordering, the algorithm of Gilmore and Gomory (1964) can be implemented

in $O(n)$ time. Step 2 of **GG** also runs in $O(n)$ time. Thus the overall complexity of **MinCycle** is $O(n^4)$ time. This completes the proof of Theorem 1. $\square$

In order to test its performance, **MinCycle** was coded in the C language on a Decstation 5000 model 200, with a speed of 32 MIPS. In generating random problem instances, we consider the following data as fixed: $\epsilon_i = 1.0$, $i = 1, \dots, 6$, $\delta_0 = 4.0$, $\delta_1 = 2.0$, $\delta_2 = 5.0$, $\delta_3 = 2.0$, $\gamma = 1.0$, $\rho = 20$, and $\mu = 28$. Thus, $a'_i = a_i + 5.0$ and $b'_i = b_i + 5.0$, $i = 1, \dots, n$. The processing times are generated independently from the uniform distribution over the integers $0, \dots, n \max$, where $n \max$ has values of 25, 50 and 75. The results in Table II show that **MinCycle** is effective at solving instances of the problem with up to 50 parts, in that all 50 instances in each row are solved optimally, as shown in the third column. The last column of Table II shows the mean CPU time in seconds required by **MinCycle** over the 50 randomly generated problem instances in each row. Computation time increases with the number of parts, but decreases slightly as processing times increase.

Algorithm **MinCycle** has been implemented in a two machine mobile-robot cell at the Manufacturing Systems Laboratory of the Industrial Engineering Department of the University of Toronto. The details of the implementation are provided by Chan and Lau (1994).

3. THREE-MACHINE CELL, SINGLE PART-TYPE

One-unit cycles are easy to find and operate, because of their simple, repetitive nature, and are thus of considerable practical importance. An important conjecture by Sethi et al. (1992) suggests that, in a robot-centered cell with equal travel times, the repetition of one-unit cycles dominates all other policies for single part-type production. Indeed, they show that this property holds for $m = 2$. If the property holds for $m \geq 2$, then the search for the optimal robot move cycle can be restricted to the $m!$ one-unit cycles in an m machine robotic cell. This will help greatly in finding the robot move cycle that minimizes cycle time. In particular, the $O(m^3)$ time algorithm of Crama

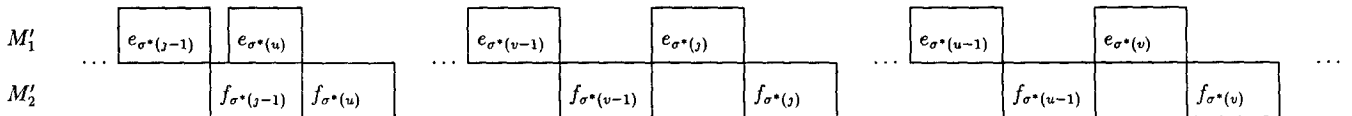


Figure 4. Partition π' and sequence $\sigma' = \{P_{\sigma^*(u)}, \sigma_{uv}^*, P_{\sigma^*(j)}, \sigma_{ju}^*, P_{\sigma^*(v)}, \sigma_{vj}^*\}$ in condition (a) of Theorem 1.

Table II
Computation Times for MinCycle

n	n max	No.	CPU (sec)
5	25	50	0.010
5	50	50	0.006
5	75	50	0.004
10	25	50	0.208
10	50	50	0.150
10	75	50	0.112
25	25	50	18.250
25	50	50	13.904
25	75	50	10.424
50	25	50	620.086
50	50	50	472.982
50	75	50	378.660

and van de Klundert (1997) will then optimize an m machine cell producing a single part-type over all possible one-unit robot move cycles. Recall that we have shown in Section 2, by a counterexample, that the conjecture does not hold for multiple part-type problems, even in a two-machine cell. For single part-type problems, we prove here that, in a very general environment, the repetition of one-unit cycles dominates more complicated policies that produce two units. The question of whether cycles producing three or more units can provide an improvement remains open, but is further partially resolved by Sriskandarajah et al. (1994) using similar methodology to that which follows.

Sethi et al. (1992) prove that there are exactly $m!$ potentially optimal one-part cycles in an m machine cell. They also show that any potentially optimal robot move cycle in an m -machine robotic cell can be described by exactly $m + 1$ basic activities:

M_i^- : Load a part on M_i , $i = 1, 2, \dots, m$,

M_m^+ : Unload a finished part from M_m .

Note that in a one-unit cycle, every basic activity must be carried out exactly once. Moreover, since in an optimal cycle we require that the robot move path is as short as possible, any two consecutive activities uniquely determine the robot moves. Therefore, a cycle can be uniquely described by a permutation of the above $m + 1$ activities. The following are the available robot move cycles for $m = 3$:

$S_1: \{M_3^+, M_1^-, M_2^-, M_3^-, M_3^+\}$,

$S_2: \{M_3^+, M_1^-, M_3^-, M_2^-, M_3^+\}$,

$S_3: \{M_3^+, M_3^-, M_1^-, M_2^-, M_3^+\}$,

$S_4: \{M_3^+, M_2^-, M_3^-, M_1^-, M_3^+\}$,

$S_5: \{M_3^+, M_2^-, M_1^-, M_3^-, M_3^+\}$,

$S_6: \{M_3^+, M_3^-, M_2^-, M_1^-, M_3^+\}$.

The environment studied here is a three-machine mobile-robot cell with unequal travel times between machines. It is important to note that, in a single part-type problem, we have $n = 1$. Thus, the cycle time is the average time to produce one part. The following preliminary result will be useful.

Lemma 6. *There are six potentially optimal cycles ($S_1, \dots, S_6$) for $MRC3|k = 1, 1 - \text{unit}|C_r$, with respective cycle times given by:*

$$T_1 = a + b + c + 4\delta_0 + 2 \sum_{i=1}^4 \delta_i + \sum_{i=1}^8 \epsilon_i - 3\gamma,$$

$$T_2 = \max \left\{ 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 4\gamma, c + 4\delta_0 + 2 \sum_{i=1}^4 \delta_i + \epsilon_1 + \epsilon_2 + \epsilon_5 + \epsilon_6 + \epsilon_7 + \epsilon_8 - 3\gamma, b + 4\delta_0 + 2\delta_2 + 2\delta_3 + \sum_{i=3}^6 \epsilon_i - \gamma, a + 4\delta_0 + 2 \sum_{i=1}^4 \delta_i + \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4 + \epsilon_7 + \epsilon_8 - 3\gamma, (a + b + c)/2 + 2\delta_0 + \delta_1 + 2\delta_2 + \delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i/2 - 3\gamma/2 \right\},$$

$$T_3 = \max \left\{ a + 6\delta_0 + 2\delta_1 + 2\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 3\gamma, c + 2\delta_0 + 2\delta_3 + 2\delta_4 + \sum_{i=5}^8 \epsilon_i - \gamma, a + b + 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + \sum_{i=1}^6 \epsilon_i - 2\gamma \right\},$$

$$T_4 = \max \left\{ b + 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 5\gamma, b + c + 4\delta_0 + 2\delta_2 + 2\delta_3 + 2\delta_4 + \sum_{i=3}^8 \epsilon_i - 2\gamma, a + b + 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + \sum_{i=1}^6 \epsilon_i - 2\gamma \right\},$$

$$T_5 = \max \left\{ c + 6\delta_0 + 2\delta_1 + 4\delta_2 + 2\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 3\gamma, b + c + 4\delta_0 + 2\delta_2 + 2\delta_3 + 2\delta_4 + \sum_{i=3}^8 \epsilon_i - 2\gamma, a + 2\delta_0 + 2\delta_1 + 2\delta_2 + \sum_{i=1}^4 \epsilon_i - \gamma \right\},$$

$$T_6 = \max \left\{ 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 4\gamma, c + 2\delta_0 + 2\delta_3 + 2\delta_4 + \sum_{i=5}^8 \epsilon_i - \gamma, b + 4\delta_0 + 2\delta_2 + 2\delta_3 + \sum_{i=3}^6 \epsilon_i - \gamma, a + 2\delta_0 + 2\delta_1 + 2\delta_2 + \sum_{i=1}^4 \epsilon_i - \gamma \right\}.$$

Proof. The cycle times given by cycles $(S_1, \dots, S_6)$ are evaluated in Appendix A. $\square$

We can now state our main result about the dominance of one-unit cycles. We need to point out that more work is necessary on this important issue. For single part-type problems, this issue needs to be addressed for cells with more than three machines. For multiple part-type problems, while we have shown in Section 2 that more complicated cycles may offer an improvement even for $m = 2$, the extent of that improvement needs to be studied.

Theorem 2. *The optimal cycle time in $MRC3|k = 1, 1 - \text{unit}|C_r$ is no larger than that in $MRC3|k = 1, \leq 2 - \text{units}|C_r$.*

Proof. We prove the theorem by using an optimization model that compares the smallest cycle time one-unit cycle with the smallest cycle time two-unit cycle. This model examines all possible data sets in a search for a counterexample to the theorem. Finding such a counterexample will disprove the theorem, but failure to do so will prove the theorem. In order to formulate the optimization model, we begin by developing the cycle time expressions for all possible two-unit cycles. As mentioned above, Sethi et al. (1992) show that any one-unit robot move cycle in an m -machine robotic cell can be described by exactly $m + 1$ basic activities. While in a one-unit cycle every basic activity must be performed exactly once, in a two-unit cycle every basic activity must be performed exactly twice. To see how many two-unit cycles are available, we characterize each cycle initially by its sequence of loadings on the machines. We let S_j^2 denote the j th two-unit cycle which we describe, and $T_j(2)$ its cycle time.

Assuming without loss of generality that a cycle starts by loading machine M_3 , there are eight possible sequences of machine loadings on M_1 and M_2 before the next loading on machine M_3 : (a) M_1^- , (b) M_2^- , (c) $M_1^- M_2^-$, (d) $M_2^- M_1^-$, (e) $M_1^- M_2^- M_1^-$, (f) $M_2^- M_1^- M_2^-$, (g) $M_1^- M_2^- M_1^- M_2^-$, and (h) $M_2^- M_1^- M_2^- M_1^-$.

Note that, in any cycle, it is not feasible to have consecutive loadings on the same machine. For example, the sequences $M_1^- M_1^-$ and $M_2^- M_2^-$ are not feasible, since between the two loadings there must be a loading on a downstream machine. Also, the sequence $M_3^- M_3^-$ would have to be followed by the sequence (g) or (h), neither of which is feasible because, after the first M_3^- , M_2 is free. Therefore, an M_2^- must precede the second M_3^- .

This gives five possible sequences of machine loadings.

1: $\{M_3^-, M_1^-, M_2^-, M_3^-, M_1^-, M_2^-\}$: (generated from (c)): see cycles S_1^2 , S_2^2 and S_3^2 below.

2: $\{M_3^-, M_1^-, M_2^-, M_3^-, M_2^-, M_1^-\}$: (generated from (c) and (d)): no feasible cycles exist, because no part is available for loading on M_2 after the second M_3^- .

3: $\{M_3^-, M_2^-, M_1^-, M_3^-, M_2^-, M_1^-\}$: (generated from (d)): see cycles S_4^2 , S_5^2 , and S_6^2 below.

4: $\{M_3^-, M_1^-, M_2^-, M_1^-, M_3^-, M_2^-\}$: (generated from (b) and (e)): see cycles $S_7^2, \dots, S_{14}^2$ below.

5: $\{M_3^-, M_2^-, M_1^-, M_2^-, M_3^-, M_1^-\}$: (generated from (a) and (f)): no feasible cycles exist, because two parts would be on M_2 simultaneously after the second M_2^- .

Now we need to insert the M_3^+ activities into the above sequences, while satisfying the condition that there is exactly one M_3^- between two consecutive M_3^+ 's. By enumerating all possible ways of inserting these activities, we derive all the 14 two-unit cycles described below.

The first loading sequence above includes a repeated partial sequence of machine loadings M_1^-, M_2^-, M_3^- . There is only one cycle (S_1^2) with one machine loading M_3^- between the two M_3^+ activities, and there are two cycles (S_2^2 and S_3^2) with two machine loadings between the two M_3^+ activities. When three machine loadings occur between the two M_3^+ activities, all possible cycles generated from this sequence are one-unit cycles, and they are not considered. Similarly, the third sequence above generates three cycles, and the fourth sequence generates eight cycles. A brief description of all the two-unit cycles now follows, with a detailed derivation for $T_{14}(2)$ given as an example.

$$S_1^2 = \{M_3^+, M_3^-, M_3^+, M_1^-, M_2^-, M_3^-, M_1^-, M_2^-, M_3^+\},$$

$$S_2^2 = \{M_3^+, M_2^-, M_3^-, M_3^+, M_1^-, M_2^-, M_3^-, M_1^-, M_3^+\},$$

$$S_3^2 = \{M_3^+, M_3^-, M_1^-, M_3^+, M_2^-, M_3^-, M_1^-, M_2^-, M_3^+\},$$

$$S_4^2 = \{M_3^+, M_3^-, M_3^+, M_2^-, M_1^-, M_3^-, M_2^-, M_1^-, M_3^+\},$$

$$S_5^2 = \{M_3^+, M_3^-, M_2^-, M_3^+, M_1^-, M_3^-, M_2^-, M_1^-, M_3^+\},$$

and $T_5(2) \geq 2T_6$, so S_5^2 is not considered further.

$$S_6^2 = \{M_3^+, M_1^-, M_3^-, M_3^+, M_2^-, M_1^-, M_3^-, M_2^-, M_3^+\},$$

$$S_7^2 = \{M_3^+, M_3^-, M_3^+, M_2^-, M_3^-, M_1^-, M_2^-, M_1^-, M_3^+\},$$

$$S_8^2 = \{M_3^+, M_3^-, M_3^+, M_1^-, M_2^-, M_1^-, M_3^-, M_2^-, M_3^+\},$$

$$S_9^2 = \{M_3^+, M_2^-, M_3^-, M_3^+, M_1^-, M_2^-, M_1^-, M_3^-, M_3^+\},$$

$$S_{10}^2 = \{M_3^+, M_1^-, M_3^-, M_3^+, M_2^-, M_3^-, M_1^-, M_2^-, M_3^+\},$$

$$S_{11}^2 = \{M_3^+, M_3^-, M_2^-, M_3^+, M_3^-, M_1^-, M_2^-, M_1^-, M_3^+\},$$

$$S_{12}^2 = \{M_3^+, M_3^-, M_1^-, M_3^+, M_2^-, M_1^-, M_3^-, M_2^-, M_3^+\},$$

and $T_{12}(2) \geq 2T_6$, so S_{12}^2 is not considered further.

$$S_{13}^2 = \{M_3^+, M_2^-, M_1^-, M_3^-, M_3^+, M_2^-, M_3^-, M_1^-, M_3^+\},$$

$$S_{14}^2 = \{M_3^+, M_1^-, M_3^-, M_2^-, M_3^+, M_3^-, M_1^-, M_2^-, M_3^+\}$$

$$\Rightarrow T_{14}(2) = a + 14\delta_0 + 4\delta_1 + 6\delta_2 + 8\delta_3 + 4\delta_4$$

$$+ 2 \sum_{i=1}^8 \epsilon_i - 7\gamma + w_1^3(14) + w_2^3(14) + w_3^3(14)$$

$$+ w_3^3(14) + w_3^3(14),$$

where

$$\begin{aligned}
w_1^3(14) &= \max\{0, a - 4\delta_0 - 2\delta_2 - 2\delta_3 - \epsilon_5 - \epsilon_6 + \gamma - w_2^2(14)\}, \\
w_2^2(14) &= \max\left\{0, b - 4\delta_0 - 2 \sum_{i=1}^4 \delta_i - \epsilon_1 - \epsilon_2 - \epsilon_7 - \epsilon_8 \right. \\
&\quad \left. + 3\gamma - w_3^1(14)\right\}, w_3^1(14) \\
&= \max\{0, b - 2\delta_0 - 2\delta_3 - 2\delta_4 - \epsilon_7 - \epsilon_8 + \gamma - w_3^2(14)\}, \\
w_3^2(14) &= \max\{0, c - 4\delta_0 - 2\delta_2 - 2\delta_3 - \epsilon_3 - \epsilon_4 + \gamma \\
&\quad - w_1^3(14)\}
\end{aligned}$$

and

$$\begin{aligned}
w_3^1(14) &= \max\left\{0, c - a - 4\delta_0 - 2\delta_1 - 2\delta_2 - 2\delta_3 \right. \\
&\quad \left. - \sum_{i=1}^4 \epsilon_i + 2\gamma\right\},
\end{aligned}$$

and where $w_i^j(q)$ denotes the waiting time of the robot before unloading part j from machine M_i using robot move cycle S_q^2 .

After deriving the equations for $T_1(2), \dots, T_{14}(2)$, we need to find out whether there exists a data set in which a two-unit cycle provides a cycle time smaller than the best cycle time available from the six potentially optimal one-unit cycles, $S_1, \dots, S_6$. To do so, we solve the following mixed integer program [MUC]. In this formulation, V is a variable which represents the difference between the smallest one-unit cycle time and the smallest average cycle time ($U/2$) for a two-unit cycle. We use $\{\text{LB}^2(j)\}$ to denote the lower bound on the two-unit cycle time $T_j(2)$. For example, $\text{LB}^2(14) = a + 14\delta_0 + 4\delta_1 + 6\delta_2 + 8\delta_3 + 4\delta_4 + 2 \sum_{i=1}^8 \epsilon_i - 7\gamma + w_1^3(14) + w_2^2(14) + w_3^2(14) + w_3^1(14)$. We also include the following constraints:

$$\begin{aligned}
w_1^3(14) &\geq a - 4\delta_0 - 2\delta_2 - 2\delta_3 - \epsilon_5 - \epsilon_6 + \gamma - w_2^2(14), \\
w_2^2(14) &\geq b - 4\delta_0 - 2 \sum_{i=1}^4 \delta_i - \epsilon_1 - \epsilon_2 - \epsilon_7 - \epsilon_8 + 3\gamma \\
&\quad - w_3^1(14), \\
w_3^2(14) &\geq b - 2\delta_0 - 2\delta_3 - 2\delta_4 - \epsilon_7 - \epsilon_8 + \gamma - w_3^1(14), \\
w_3^1(14) &\geq c - 4\delta_0 - 2\delta_2 - 2\delta_3 - \epsilon_3 - \epsilon_4 + \gamma - w_1^3(14), \\
w_3^1(14) &\geq c - a - 4\delta_0 - 2\delta_1 - 2\delta_2 - 2\delta_3 - \sum_{i=1}^4 \epsilon_i + 2\gamma.
\end{aligned}$$

Finally, we let $\text{LB}_i^1(j)$ denote the i th lower bound in the expression for T_j in Lemma 6.

Maximize V

subject to: $V \leq T_j - U/2, \quad j = 1, \dots, 6,$

$U \geq T_j(2) - MY_j, \quad j = 1, \dots, 4, 6, \dots, 11, 13, 14,$

$$\sum_{j \neq 5, 12}^{14} Y_j = 11,$$

$$T_j(2) \geq \{\text{LB}^2(j)\}, \quad j = 1, \dots, 4, 6, \dots, 11, 13, 14,$$

$$T_1 = a + b + c + 4\delta_0 + 2 \sum_{i=1}^4 \delta_i + \sum_{i=1}^8 \epsilon_i - 3\gamma,$$

$$T_2 \leq \text{LB}_j^1(2) + M(1 - W_{2j}), \quad j = 1, \dots, 5,$$

$$\sum_{j=1}^4 W_{2j} = 1,$$

$$T_3 \leq \text{LB}_j^1(3) + M(1 - W_{3j}), \quad j = 1, \dots, 3,$$

$$\sum_{j=1}^3 W_{3j} = 1,$$

$$T_4 \leq \text{LB}_j^1(4) + M(1 - W_{4j}), \quad j = 1, \dots, 3,$$

$$\sum_{j=1}^3 W_{4j} = 1,$$

$$T_5 \leq \text{LB}_j^1(5) + M(1 - W_{5j}), \quad j = 1, \dots, 3,$$

$$\sum_{j=1}^3 W_{5j} = 1$$

$$T_6 \leq \text{LB}_j^1(6) + M(1 - W_{6j}), \quad j = 1, \dots, 4,$$

$$\sum_{j=1}^4 W_{6j} = 1,$$

$$\gamma \leq \delta_j, \quad j = 1, \dots, 4,$$

$$a, b, c, \delta_0, \dots, \delta_4, \epsilon_1, \dots, \epsilon_8, \gamma, T_1, \dots, T_6,$$

$$T_1(2), \dots, T_4(2), T_6(2), \dots, T_{11}(2), T_{13}(2),$$

$$T_{14}(2), U, V, w_i^j(q) \geq 0, \quad W_{ij}, Y_j \in \{0, 1\},$$

where M is a large, positive number. Including the 44 $w_i^j(q)$ variables, the model has 81 continuous and 30 zero-one variables, and 103 constraints. We note, from the first constraint, that [MUC] attempts to maximize the difference between the quantities $\min_{1 \leq j \leq 6} \{T_j\}$ and $U/2$. Now, $\min_{1 \leq j \leq 6} \{T_j\}$ is the cycle time to produce one unit by using the best of the one-unit cycles. Also, from the second and third constraints, $U/2$ is the average cycle time using the best of the two-unit cycles. It is thus easy to see that there exists a data set in which $\min_{1 \leq j \leq 6} \{T_j\} > \min_{1 \leq j \neq 5, 12 \leq 14} \{T_j(2)\}/2$, where $T_j(2)$ is taken as the maximum of its lower bound components, *if and only if* there exists an optimal solution to program [MUC] with $V > 0$. Program [MUC] was solved using the software package GAMS, and $V = 0$ in the optimal solution. This completes the proof of Theorem 2. $\square$

Because the number of cycles that produce three or more units is large, we have not enumerated them. Of course, the methodology we describe here could be used to check whether such cycles could provide an improvement in cycle time. In view of the simplicity of one-unit cycles and the results in this section which provide easy comparability of their cycle times, we recommend their use.

4. THREE-MACHINE CELL, MULTIPLE PART-TYPES

In this section, we study the multiple part-types problem in a three machine mobile-robot cell, or $MRC3|k \geq 2|C_t$. Sethi et al. (1992) show that there are six potentially optimal one-unit robot move cycles ($S_1, \dots, S_6$) to consider in the three-machine cell. These cycles are determined by the possible robot activities, rather than by the cell data, so this result continues to hold when we make a variety of assumptions about the cell data. It is important to investigate the complexity of the part sequencing problem with one-unit cycles, since those cycles offer the most easily understood and implemented ways of moving the robot around the cell. We will show, in Section 4.1, that the optimal part sequencing problems associated with four of those six cycles are polynomially solvable. However, in the other two cycles, the recognition version of the part sequencing problem is unary NP -complete (Hall et al. 1997). This enables us to identify several special cases for which the problem is polynomially solvable. We then study, in Section 4.2, the ways in which a robotic cell converges to a steady state.

4.1. Efficiently Solvable Special Cases

We now study the cycle time minimization problem under robot move cycles S_1, S_3, S_4 , and S_5 . Where appropriate, we let $a_{\sigma(n+1)} = a_{\sigma(1)}$, $a_{\sigma(n+2)} = a_{\sigma(2)}$, and similarly for $b_{\sigma(n+1)}$, $b_{\sigma(n+2)}$, $c_{\sigma(n+1)}$ and $c_{\sigma(n+2)}$. We also let $w_i^{n+1} = w_i^1$ for $i = 1, 2, 3$. We let $T_{j\sigma(t)\sigma(t+1)}^h$ denote the time between the loading of part $P_{\sigma(t)}$ on machine M_h , and the loading of part $P_{\sigma(t+1)}$ on M_h using robot move cycle S_j , where $h \in \{1, 2, 3\}$ and $j \in \{1, \dots, 6\}$.

Theorem 3. *Problem $MRC3|k \geq 2, S_1|C_t$ can be solved trivially.*

Proof. From Appendix A, we have

$$T_{1(\sigma)} = \sum_{i=1}^n T_{1\sigma(i)\sigma(i+1)}^1 = n\alpha_1 + \sum_{i=1}^n (a_i + b_i + c_i).$$

Clearly, $T_{1(\sigma)}$ depends only on $a_i, b_i, c_i, 1 \leq i \leq n, \alpha_1$ and n , and is independent of the sequence of parts. $\square$

In Theorems 4, 5, and 6, we show that the part sequencing problem under robot move cycles S_3, S_4 and S_5 , respectively, can be reduced to a problem of the form: minimize $(C + \sum_{i=1}^n \max\{e_{\sigma(i+1)}, f_{\sigma(i)}\})$, for some constant C . Any such problem can be solved optimally in $O(n \log n)$ time by the algorithm of Gilmore and Gomory (1964).

Theorem 4. *Problem $MRC3|k \geq 2, S_3|C_t$ can be solved optimally in $O(n \log n)$ time.*

Proof. From Appendix A, we have

$$\begin{aligned} T_{3\sigma(i)\sigma(i+1)}^3 \\ = \alpha_3 + \max\{\beta_3 + a_{\sigma(i+1)}, b_{3\sigma(i+1)} + a_{\sigma(i+1)}, c_{3\sigma(i)}\}. \end{aligned}$$

Letting $e_{3\sigma(i)} = \max\{\beta_3 + a_{\sigma(i)}, b_{3\sigma(i)} + a_{\sigma(i)}\}$, we have

$$T_{3(\sigma)} = \sum_{i=1}^n T_{3\sigma(i)\sigma(i+1)}^3 = n\alpha_3 + \sum_{i=1}^n \max\{e_{3\sigma(i+1)}, c_{3\sigma(i)}\}. \quad \square$$

Theorem 5. *Problem $MRC3|k \geq 2, S_4|C_t$ can be solved optimally in $O(n \log n)$ time.*

Proof. From Appendix A, we have

$$\begin{aligned} T_{4\sigma(i+1)\sigma(i+2)}^1 \\ = \alpha_4 + b_{\sigma(i+1)} + \max\{\beta_4, c_{4\sigma(i)}, a_{4\sigma(i+1)}\}. \end{aligned}$$

Letting $e_{4\sigma(i)} = \max\{\beta_4, a_{4\sigma(i)}\}$, we have

$$\begin{aligned} T_{4(\sigma)} &= \sum_{i=1}^n T_{4\sigma(i+1)\sigma(i+2)}^1 \\ &= n\alpha_4 + \sum_{i=1}^n b_{\sigma(i)} + \sum_{i=1}^n \max\{e_{4\sigma(i+1)}, c_{4\sigma(i)}\}. \quad \square \end{aligned}$$

Theorem 6. *Problem $MRC3|k \geq 2, S_5|C_t$ can be solved optimally in $O(n \log n)$ time.*

Proof. From Appendix A, we have

$$\begin{aligned} T_{5\sigma(i+1)\sigma(i+2)}^1 \\ = \alpha_5 + \max\{\beta_5 + c_{\sigma(i)}, b_{5\sigma(i)} + c_{\sigma(i)}, a_{5\sigma(i+1)}\}. \end{aligned}$$

Letting $f_{5\sigma(i)} = \max\{\beta_5 + c_{\sigma(i)}, b_{5\sigma(i)} + c_{\sigma(i)}\}$, we have

$$\begin{aligned} T_{5(\sigma)} &= \sum_{i=1}^n T_{5\sigma(i+1)\sigma(i+2)}^1 \\ &= n\alpha_5 + \sum_{i=1}^n \max\{a_{5\sigma(i+1)}, f_{5\sigma(i)}\}. \quad \square \end{aligned}$$

Let $T_{j(\sigma)}$ denote the cycle time for robot move cycle S_j , $j = 1, \dots, 6$, in $MRC3|k \geq 2, 1 - \text{unit}|C_t$ using part sequence σ . In order to derive some conditions under which this problem is polynomially solvable, we need the following preliminary result.

We begin by characterizing the production cycle time for parts $P_{\sigma(1)}, \dots, P_{\sigma(n)}$ under robot move cycle S_2 , where σ denotes the sequence of parts.

Lemma 7. *The cycle time for the production of n parts in sequence σ under cycle S_2 is given by*

$$\begin{aligned} T_{2(\sigma)} &= n\alpha_2 + \sum_{i=1}^n \max\{\beta_2, a_{2\sigma(i+2)}, b_{2\sigma(i+1)} - w_3^i\} \\ &\quad + \sum_{i=1}^n w_3^i, \end{aligned}$$

where $w_3^i = \max\{0, c_{2\sigma(i)} - \max\{\beta_2, a_{2\sigma(i+1)} - \max\{0, b_{2\sigma(i)} - \beta_2 - w_3^{i-1}\}\}\}$.

Proof. See Appendix A. $\square$

Remark 2. When $\delta_0 = 0$, $\delta_i = \delta, i = 1, \dots, 4$, $\epsilon_i = \epsilon, i = 1, \dots, 8$, and $\gamma = 0$, the cell is robot-centered. This problem is denoted $RCC3|k \geq 2, S_2, \delta_i = \delta, \epsilon_i = \epsilon, \gamma = 0|C_t$.

From Lemma 7, the cycle time for the production of n parts is given by

$$T_{2(\sigma)} = n\alpha + \bar{T}_\sigma, \quad \text{where}$$

$$\bar{T}_\sigma = \sum_{i=1}^n \max\{\beta, a_{\sigma(i+2)} + \beta/2, b_{\sigma(i+1)} - w_3^i\} + \sum_{i=1}^n w_3^i,$$

$$w_3^i = \max\{0, c_{\sigma(i)} - \max\{\beta/2, a_{\sigma(i+1)} - \max\{0, b_{\sigma(i)} - \beta - w_3^{i-1}\}\}\},$$

$$\alpha = 4\delta + 4\epsilon, \text{ and } \beta = 8\delta + 4\epsilon.$$

We next characterize the production cycle time for robot move cycle S_6 .

Lemma 8. *The cycle time for the production of n parts in sequence σ under cycle S_6 is given by*

$$T_{6(\sigma)} = n\alpha_6 + \sum_{i=1}^n \max\{\beta_6, c_{6\sigma(i)} - w_1^{i+1}, b_{6\sigma(i+1)}, a_{6\sigma(i+2)}\},$$

$$\text{where } w_1^{i+1} = \max\{0, a_{6\sigma(i+1)} - \max\{\beta_6, b_{6\sigma(i)}, c_{6\sigma(i-1)} - w_1^i\}\}.$$

Proof. See Appendix A. $\square$

Remark 3. Similar to Remark 2, we consider problem RCC3| $k \geq 2$, S_6 , $\delta_i = \delta$, $\epsilon_i = \epsilon$, $\gamma = 0|C_r$. From Lemma 8, the cycle time for the production of n parts is given by

$$T_{6(\sigma)} = n\alpha + \bar{T}_\sigma, \quad \text{where}$$

$$\bar{T}_\sigma = \sum_{i=1}^n \max\{\beta, c_{\sigma(i)} - w_1^{i+1}, b_{\sigma(i+1)}, a_{\sigma(i+2)}\},$$

where

$$w_1^{i+1} = \max\{0, a_{\sigma(i+1)} - \max\{\beta, b_{\sigma(i)}, c_{\sigma(i-1)} - w_1^i\}\},$$

$$\alpha = 4\delta + 4\epsilon, \text{ and } \beta = 8\delta + 4\epsilon.$$

The next result will be useful for the theorem that follows.

Lemma 9. *In MRC3| $k \geq 2$, $1 - \text{unit}|C_r$,*

$$T_{2(\sigma)}, T_{6(\sigma)} \geq n \left(8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 4\gamma \right).$$

Proof. Using S_2 , as described in Appendix A, $T_{2(\sigma)} \geq n(\alpha_2 + \beta_2) = n(8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 4\gamma)$.

Similarly, using S_6 , as described in Appendix A, $T_{6(\sigma)} \geq n(\alpha_6 + \beta_6) = n(8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 4\gamma)$. $\square$

It follows from Theorems 3, 4, 5, and 6 that problem MRC3| $k \geq 2$, $1 - \text{unit}|C_r$ is polynomially solvable in two situations. The first of these is when the data is such that the best sequences under S_2 and S_6 can be found in polynomial time. The second situation is when the cycle time provided by the best sequence under S_2 or S_6 cannot be optimal. In the theorem that follows, condition A corresponds to the first situation, and conditions B through E to the second.

Theorem 7. *Problem MRC3| $k \geq 2$, $1 - \text{unit}|C_r$ can be solved optimally in $O(n \log n)$ time under any of the conditions A, B, C, D, or E below:*

$$\text{A: } \delta_i = \delta, i = 0, \dots, 4, \epsilon_i = \epsilon, i = 1, \dots, 8,$$

$$\gamma = 0 \text{ and } c_i \leq 4\delta + 2\epsilon, i = 1, \dots, n,$$

$$\text{B: } \sum_{i=1}^n (a_i + b_i + c_i) \leq n(4\delta_0 + 2\delta_2 + 2\delta_3 - \gamma),$$

$$\text{C1: } a_i \leq 2\delta_0 + 2\delta_2 - \gamma, i = 1, \dots, n,$$

$$\text{C2: } b_i \leq 2\delta_0 + 2\delta_3 + 2\delta_4 + \epsilon_7 + \epsilon_8 - \gamma, \\ i = 1, \dots, n,$$

$$\text{C3: } c_i \leq 6\delta_0 + 2\delta_1 + 4\delta_2 + 2\delta_3 + \epsilon_1 + \epsilon_2 + \epsilon_3 \\ + \epsilon_4 - 3\gamma, i = 1, \dots, n,$$

$$\text{D1: } a_i \leq 4\delta_0 + 2\delta_2 + 2\delta_3 + 2\delta_4 + \epsilon_7 + \epsilon_8 - 3\gamma, \\ i = 1, \dots, n,$$

$$\text{D2: } b_i \leq \gamma, i = 1, \dots, n,$$

$$\text{D3: } c_i \leq 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + \epsilon_1 + \epsilon_2 - 3\gamma, \\ i = 1, \dots, n,$$

$$\text{E1: } a_i \leq 6\delta_0 + 2\delta_2 + 4\delta_3 + 2\delta_4 + \epsilon_5 + \epsilon_6 + \epsilon_7 \\ + \epsilon_8 - 3\gamma, i = 1, \dots, n,$$

$$\text{E2: } b_i \leq 2\delta_0 + 2\delta_1 + 2\delta_2 + \epsilon_1 + \epsilon_2 - \gamma, \\ i = 1, \dots, n,$$

$$\text{E3: } c_i \leq 2\delta_0 + 2\delta_3 - \gamma, i = 1, \dots, n.$$

Proof. We consider each condition A through E in turn.

A: From Remark 2 and the conditions, $T_{2(\sigma)} = n(4\delta + 4\epsilon) + \sum_{i=1}^n \max\{8\delta + 4\epsilon, a_{\sigma(i+2)} + 4\delta + 2\epsilon, b_{\sigma(i+1)}\}$, where if we let $f_{\sigma(i)} = \max\{8\delta + 4\epsilon, a_{\sigma(i+2)} + 4\delta + 2\epsilon\}$, the part sequencing problem can be solved by the algorithm of Gilmore and Gomory (1964) in $O(n \log n)$ time. Similarly, from Remark 3 and the conditions, we have $T_{6(\sigma)} = n(4\delta + 4\epsilon) + \sum_{i=1}^n \max\{8\delta + 4\epsilon, b_{\sigma(i+1)}, a_{\sigma(i+2)}\}$, with the same result. It then follows from Theorems 3, 4, 5, and 6 that the optimal part sequences under all six one-unit cycles can be found, and their cycle times compared, in $O(n \log n)$ time.

B: $T_{1(\sigma)} = \sum_{i=1}^n (a_i + b_i + c_i) + n(4\delta_0 + 2\sum_{i=1}^4 \delta_i + \sum_{i=1}^8 \epsilon_i - 3\gamma) \leq n(4\delta_0 + 2\delta_2 + 2\delta_3 - \gamma) + n(4\delta_0 + 2\sum_{i=1}^4 \delta_i + \sum_{i=1}^8 \epsilon_i - 3\gamma)$ from condition B, $\leq T_{2(\sigma)}, T_{6(\sigma)}$ from Lemma 9.

C: From Theorem 4, $T_{3(\sigma)} = \sum_{i=1}^n \max\{\alpha_3 + \beta_3 + a_{\sigma(i+1)}, \alpha_3 + b_{3\sigma(i+1)} + a_{\sigma(i+1)}, \alpha_3 + c_{3\sigma(i)}\} \leq \sum_{i=1}^n \max\{(2\delta_0 + 2\delta_3 + \epsilon_5 + \epsilon_6 - \gamma) + (4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + 2\delta_4 + \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4 + \epsilon_7 + \epsilon_8 - 2\gamma) + (2\delta_0 + 2\delta_2 - \gamma), (2\delta_0 + 2\delta_3 + \epsilon_5 + \epsilon_6 - \gamma) + (2\delta_0 + 2\delta_3 + 2\delta_4 + \epsilon_7 + \epsilon_8 - \gamma + 2\delta_0 + 2\delta_1 + 2\delta_2 + \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4 - \gamma) + (2\delta_0 + 2\delta_2 - \gamma), (2\delta_0 + 2\delta_3 + \epsilon_5 + \epsilon_6 - \gamma) + (6\delta_0 + 2\delta_1 + 4\delta_2 + 2\delta_3 + \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4 - 3\gamma + 2\delta_4 + \epsilon_7 + \epsilon_8)\} = \sum_{i=1}^n \max\{8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 4\gamma, 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 4\gamma, 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 4\gamma\} \leq T_{2(\sigma)}, T_{6(\sigma)}$ from Lemma 9.

D: The proof is similar to part C above, using Theorem 5.

E: The proof is similar to part C above, using Theorem 6. $\square$

Remark 4. The results in Theorem 7 provide considerable intuition about tradeoffs between different robot move cycles. Specifically, each of the conditions in Theorem 7 suggests that, if processing times are short relative to robot travel times, it will be better to wait at a machine during processing as cycles S_1 , S_3 , S_4 , and S_5 usually do, rather than drop the part and move elsewhere, as cycles S_2 and S_6 do.

4.2. Steady State Analysis

All the results in the paper relate to the performance of robotic cells operating in a steady state, as described in Definition 1 below. Thus, a question to be asked is whether the cell must be initialized in its steady state. Alternatively, what happens if it starts from a state which is not part of its steady state cycle? We show here that the cell reaches a steady state in a number of cycles which is bounded by a function of the cell data. Usually the production of parts begins from an initial state $\hat{E} = (\phi, \phi, \phi, I)$, where the cell is empty and the robot is at I ready to pick up a part. We first show how a cell converges to a steady state starting from state $\hat{E}_0$, under robot move cycles $S_1, \dots, S_6$ in MRC3. Recall that the corresponding cycle time expressions are derived in Appendix A.

Definition 1. A steady state is a condition within a cell in which the waiting time of the robot, for each part on each machine, is identical to its value in the previous MPS.

In robot move cycles S_1 , S_3 , S_4 , and S_5 , there is a state E^* in which the robot has just finished loading a part on a machine and all other machines are free. In S_1 that machine can be M_1 , M_2 , or M_3 ; in S_3 the machine is M_3 ; in S_4 the machine can be M_2 or M_3 ; and in S_5 the machine is M_2 . In every case, E^* can be reached from the state $\hat{E}$ within the production of one MPS. Therefore, the process of convergence to a steady state is very simple in those four cycles. For the same reason, the two cycles S_1 and S_2 in MRC2 converge to a steady state within the production of one MPS, from the initial starting state $\hat{E}$.

Remark 5. In MRC3, the state E^* effectively decomposes production into MPSs in cycles S_1 , S_3 , S_4 , and S_5 . Thus, a single MPS may be analyzed by itself. This is one of the properties that makes the part sequencing problem using those cycles solvable in polynomial time, as discussed in Section 4.1.

It remains to consider the process of convergence to a steady state using robot move cycles S_2 and S_6 . We first consider cycle S_2 , for which the cycle time expression is derived in Lemma 7. We need two preliminary results.

Lemma 10. Let $w_3^1, \dots, w_3^n$ and $w_3'^1, \dots, w_3'^n$ denote two feasible vectors of waiting times in S_2 , where $w_3^{t+1} = \max\{0, c_{2\sigma(t+1)} - \max\{\beta_2, a_{2\sigma(t+2)} - \max\{0, b_{2\sigma(t+1)} - w_3^t - \beta_2\}\}\}$, and $w_3'^{t+1} = \max\{0, c_{2\sigma(t+1)} - \max\{\beta_2, a_{2\sigma(t+2)} - \max\{0, b_{2\sigma(t+1)} - w_3'^t - \beta_2\}\}\}$. If $w_3^t \leq w_3'^t$, then $w_3^{t+1} \leq w_3'^{t+1}$ and $w_3^{t+2} \leq w_3'^{t+2}$.

Proof. Since $w_3'^t \geq w_3^t$,

$$\begin{aligned} & \max\{0, b_{2\sigma(t+1)} - w_3'^t - \beta_2\} \\ & \leq \max\{0, b_{2\sigma(t+1)} - w_3^t - \beta_2\} \\ & \Rightarrow \max\{\beta_2, a_{2\sigma(t+2)} \\ & \quad - \max\{0, b_{2\sigma(t+1)} - w_3'^t - \beta_2\}\} \\ & \geq \max\{\beta_2, a_{2\sigma(t+2)} \\ & \quad - \max\{0, b_{2\sigma(t+1)} - w_3^t - \beta_2\}\} \\ & \Rightarrow \max\{0, c_{2\sigma(t+1)} \\ & \quad - \max\{\beta_2, a_{2\sigma(t+2)} \\ & \quad - \max\{0, b_{2\sigma(t+1)} - w_3'^t - \beta_2\}\}\} \\ & \leq \max\{0, c_{2\sigma(t+1)} \\ & \quad - \max\{\beta_2, a_{2\sigma(t+2)} \\ & \quad - \max\{0, b_{2\sigma(t+1)} - w_3^t - \beta_2\}\}\} \\ & \Rightarrow w_3'^{t+1} \leq w_3^{t+1}. \end{aligned}$$

The second part of the proof follows from the first part, by setting $i = i + 1$. $\square$

Due to the behavior of the waiting times w_3^t in Lemma 10, we need the following lemma to define a steady state for the S_2 cycle.

Lemma 11. For n even, the S_2 cycle consisting of a single MPS defines a steady state. However, for n odd, two MPS's may be required to define a steady state for the S_2 cycle.

Proof. Let $w_3^t(q)$ denote the waiting time of the robot at machine M_j for part $P_{\sigma(t)}$ in the q th cycle. If n is even, it is clear from the recursive relations for $w_3^t(q)$ and Lemma 10 that, if $w_3^t(q) \leq w_3'^t(q)$, then $w_3^t(q+1) \leq w_3'^t(q+1)$. Thus, a cycle containing one MPS is sufficient to define a steady state. If n is odd, it is similarly clear that, if $w_3^t(q) \leq w_3'^t(q)$, then $w_3^t(q+1) \geq w_3'^t(q+1)$. However, $w_3^t(q+2) \leq w_3'^t(q+2)$. Therefore, two MPS's may be required to define a steady state. $\square$

In view of Lemma 11, we let v denote the number of parts in a cycle, where $v = 2n$ if n is odd, and $v = n$ if n is even. For n odd, a given part sequence σ means a sequence of $2n$ parts in which the first n parts corresponding to an MPS has the same order as the last n parts.

Hall et al. (1997) show that the problem of finding the best part sequence to be processed using robot move cycle S_2 is intractable. They also describe an $O(n^2)$ time algorithm to find the steady state waiting times and the cycle time, for a given part sequence under S_2 .

There are two different ways in which cycle S_2 can be initialized from the state $\hat{E} = (\phi, \phi, \phi, I)$ in MRC3/ $k \geq 2$, S_2/C_i :

(a) Initialization **INA2**, starting with part $P_{\sigma(i)}$: pick up $P_{\sigma(i)}: (\epsilon_1)$, move to $M_1: (\delta_1)$, load $P_{\sigma(i)}$ on $M_1: (\epsilon_2)$, wait at

$M_1:(w'_1(0) = a_{\sigma(i)})$, unload $P_{\sigma(i)}$ from $M_1:(\epsilon_3)$, move to $M_2:(\delta_0 + \delta_2)$, load $P_{\sigma(i)}$ on $M_2:(\epsilon_4)$, move to $I:(\delta_0 + \delta_1 + \delta_2 - \gamma)$, pick up $P_{\sigma(i+1)}:(\epsilon_1)$, move to $M_1:(\delta_1)$, load $P_{\sigma(i+1)}$ on $M_1:(\epsilon_2)$, move to $M_2:(\delta_0 + \delta_2)$, if necessary wait at $M_2:(w'_2(0))$, unload $P_{\sigma(i)}$ from $M_2:(\epsilon_5)$, move to $M_3:(\delta_0 + \delta_3)$, load $P_{\sigma(i)}$ on $M_3:(\epsilon_6)$, move to $M_1:(2\delta_0 + \delta_2 + \delta_3 - \gamma)$, if necessary wait at $M_1:(w'_1+1(0))$, unload $P_{\sigma(i+1)}$ from $M_1:(\epsilon_3)$, move to $M_2:(\delta_0 + \delta_2)$, and load $P_{\sigma(i+1)}$ on $M_2:(\epsilon_4)$. At this point, cycle S_2 starts from the state $E = (\phi, \Omega, \Omega, M_2^-)$.

The waiting time equations corresponding to initialization **INA2** are as follows, where $w'_j(0)$ denotes the waiting time of the robot at machine M_j for part $P_{\sigma(i)}$ in the initialization step:

$$\begin{aligned} w'_1(0) &= a_{\sigma(i)}, \\ w'_2(0) &= \max\{0, b_{\sigma(i)} - 2\delta_0 - 2\delta_1 - 2\delta_2 - \epsilon_1 - \epsilon_2 + \gamma\}, \\ w'_1+1(0) &= \max\{0, a_{\sigma(i+1)} - w'_2(0) - 4\delta_0 - 2\delta_2 - 2\delta_3 - \epsilon_5 - \epsilon_6 + \gamma\}, \\ w'_3(0) &= \max\{0, c_{\sigma(i)} - w'_1+1(0) - 4\delta_0 - 2\delta_2 - 2\delta_3 - \epsilon_3 - \epsilon_4 + \gamma\}. \end{aligned}$$

The waiting time equations corresponding to the q th S_2 cycle are as follows, where $w'_j(q)$ denotes the waiting time of the robot at machine M_j for part $P_{\sigma(i)}$ in the q th cycle, $q = 1, 2, \dots$, and we let $w'_3+1(q) = w'_3(q)$, $j = 1, \dots, v$:

$$\begin{aligned} w'_3(q) &= \max\{0, c_{2\sigma(j)} - \max\{\beta_2, a_{2\sigma(j+1)} - \max\{0, b_{2\sigma(j)} - \beta_2 - w'_3-1(q-1)\}\}\}, \\ j &= i+1, \\ w'_3(q) &= \max\{0, c_{2\sigma(j)} - \max\{\beta_2, a_{2\sigma(j+1)} - \max\{0, b_{2\sigma(j)} - \beta_2 - w'_3-1(q)\}\}\}, \\ j &= i+2, \dots, i+v, \end{aligned}$$

where $P_{\sigma(i)}$ is the first part produced, starting from $\hat{E}$.

(b) Initialization **INB2**, starting with part $P_{\sigma(i)}$: pick up $P_{\sigma(i)}:(\epsilon_1)$, move to $M_1:(\delta_1)$, load $P_{\sigma(i)}$ on $M_1:(\epsilon_2)$, wait at $M_1:(w'_1(0) = a_{\sigma(i)})$, unload $P_{\sigma(i)}$ from $M_1:(\epsilon_3)$, move to $M_2:(\delta_0 + \delta_2)$, load $P_{\sigma(i)}$ on $M_2:(\epsilon_4)$, wait at $M_2:(w'_2(0) = b_{\sigma(i)})$, unload $P_{\sigma(i)}$ from $M_2:(\epsilon_5)$, move to $M_3:(\delta_0 + \delta_3)$, load $P_{\sigma(i)}$ on $M_3:(\epsilon_6)$, move to $I:(2\delta_0 + \delta_1 + \delta_2 + \delta_3 - 2\gamma)$, pick up $P_{\sigma(i+1)}:(\epsilon_1)$, move to $M_1:(\delta_1)$, and load $P_{\sigma(i+1)}$ on $M_1:(\epsilon_2)$, wait at $M_1:(w'_1+1(0) = a_{\sigma(i+1)})$, unload $P_{\sigma(i+1)}$ from $M_1:(\epsilon_3)$, move to $M_2:(\delta_0 + \delta_2)$, and load $P_{\sigma(i+1)}$ on $M_2:(\epsilon_4)$. At this point, cycle S_2 starts from the state $E = (\phi, \Omega, \Omega, M_2^-)$.

The waiting time equations corresponding to initialization **INB2** are as follows:

$$\begin{aligned} w'_1(0) &= a_{\sigma(i)}, \\ w'_2(0) &= b_{\sigma(i)}, \\ w'_1+1(0) &= a_{\sigma(i+1)}, \\ w'_3(0) &= \max\{0, c_{\sigma(i)} - w'_1+1(0) - 4\delta_0 - 2\delta_1 - 2\delta_2 - 2\delta_3 - \epsilon_1 - \epsilon_2 - \epsilon_3 - \epsilon_4 + 2\gamma\}. \end{aligned}$$

The waiting time equations corresponding to the q th S_2 cycle are as follows, where $q = 1, 2, \dots$, and $w'_3+1(q) = w'_3(q)$, $j = 1, \dots, v$:

$$\begin{aligned} w'_3(q) &= \max\{0, c_{2\sigma(j)} - \max\{\beta_2, a_{2\sigma(j+1)} - \max\{0, b_{2\sigma(j)} - \beta_2 - w'_3-1(q-1)\}\}\}, \\ j &= i+1, \\ w'_3(q) &= \max\{0, c_{2\sigma(j)} - \max\{\beta_2, a_{2\sigma(j+1)} - \max\{0, b_{2\sigma(j)} - \beta_2 - w'_3-1(q)\}\}\}, \\ j &= i+2, \dots, i+v. \end{aligned}$$

The system can be initialized in two different ways, **INA2** and **INB2**, and in each case there are n different possible parts in σ to start with. Thus, there are $2n$ possible ways to initialize the system. The following theorem uses one of them to provide an upper bound on the number of MPS cycles needed to reach a steady state.

Theorem 8. Starting from the state $\hat{E} = (\phi, \phi, \phi, I)$, using part sequence σ , the cell in $MRC3|k \geq 2, S_2|C_t$ will go through at most $\max\{1, \min_{1 \leq i \leq v} \{s_i\}\}$ cycles before reaching a steady state, where $s_i = c_{2\sigma(i)} - \max\{\beta_2, a_{2\sigma(i+1)} - \max\{0, b_{2\sigma(i+1)} - \beta_2\}\}$, $i = 1, 2, \dots, v$, and this bound is attainable.

Proof. Assume without loss of generality that all the parameters are integer. The waiting time equations in Lemma 7 show that, once the same value of w'_3 occurs in two consecutive cycles for some i , $1 \leq i \leq v$, a steady state has been reached. An induction argument, based on Lemma 10, shows that $w'_3(q)$ monotonically increases or decreases as q increases, until a steady state is reached. We have $w'_3 = 0$ if $s_i \leq 0$, for some i , $1 \leq i \leq v$. In this case, the system reaches a steady state in the first cycle. On the other hand, we have $0 \leq w'_3 \leq s_i$ if $s_i > 0$ for $i = 1, \dots, v$. In this case, it takes at most $\min_{1 \leq i \leq v} \{s_i\}$ cycles before the system reaches a steady state.

Example 1

$$\begin{aligned} n &= 4, \\ a_{2\sigma(1)} &= X, b_{2\sigma(1)} = X, c_{2\sigma(1)} = X, \\ a_{2\sigma(2)} &= X, b_{2\sigma(2)} = X, c_{2\sigma(2)} = X + 1, \\ a_{2\sigma(3)} &= X, b_{2\sigma(3)} = X, c_{2\sigma(3)} = X, \\ a_{2\sigma(4)} &= X, b_{2\sigma(4)} = X, c_{2\sigma(4)} = X, \end{aligned}$$

where $\sigma = (P_1, P_2, P_3, P_4)$, and $X > 0$ is an integer.

$$\begin{aligned} \epsilon_i &= 0, i = 1, \dots, 8, \delta_0 = 0, \\ \delta_i &= 0, i = 1, \dots, 4, \gamma = 0, \beta_2 = 0. \end{aligned}$$

We use initialization **INA2**, starting with part $P_{\sigma(1)}$ in sequence σ . We have

$$\begin{aligned}
 w_1^1(0) &= a_{\sigma(1)} = X, \\
 w_2^1(0) &= \max\{0, b_{2\sigma(1)} - 2\delta_0 - 2\delta_1 - 2\delta_2 - \epsilon_1 - \epsilon_2 \\
 &\quad + \gamma\} = X, \\
 w_1^2(0) &= \max\{0, a_{2\sigma(2)} - w_2^1(0) - 4\delta_0 - 2\delta_2 - 2\delta_3 \\
 &\quad - \epsilon_5 - \epsilon_6 + \gamma\} = 0, \\
 w_3^1(0) &= \max\{0, c_{2\sigma(1)} - w_1^2(0) - 4\delta_0 - 2\delta_2 - 2\delta_3 \\
 &\quad - \epsilon_3 - \epsilon_4 + \gamma\} = X, \\
 w_3^2(1) &= \max\{0, c_{2\sigma(2)} - \max\{\beta_2, a_{2\sigma(3)} - \max\{0, b_{2\sigma(2)} \\
 &\quad - \beta_2 - w_3^1(0)\}\}\} = 1, \\
 w_3^3(1) &= \max\{0, c_{2\sigma(3)} - \max\{\beta_2, a_{2\sigma(4)} - \max\{0, b_{2\sigma(3)} \\
 &\quad - \beta_2 - w_3^2(1)\}\}\} = X - 1, \\
 w_3^4(1) &= \max\{0, c_{2\sigma(4)} - \max\{\beta_2, a_{2\sigma(1)} - \max\{0, b_{2\sigma(4)} \\
 &\quad - \beta_2 - w_3^3(1)\}\}\} = 1, \\
 w_3^1(1) &= \max\{0, c_{2\sigma(1)} - \max\{\beta_2, a_{2\sigma(2)} - \max\{0, b_{2\sigma(1)} \\
 &\quad - \beta_2 - w_3^4(1)\}\}\} = X - 1, \\
 w_3^2(2) &= \max\{0, c_{2\sigma(2)} - \max\{\beta_2, a_{2\sigma(3)} - \max\{0, b_{2\sigma(2)} \\
 &\quad - \beta_2 - w_3^1(1)\}\}\} = 2, \\
 w_3^3(2) &= \max\{0, c_{2\sigma(3)} - \max\{\beta_2, a_{2\sigma(4)} - \max\{0, b_{2\sigma(3)} \\
 &\quad - \beta_2 - w_3^2(2)\}\}\} = X - 2, \\
 w_3^4(2) &= \max\{0, c_{2\sigma(4)} - \max\{\beta_2, a_{2\sigma(1)} - \max\{0, b_{2\sigma(4)} \\
 &\quad - \beta_2 - w_3^3(2)\}\}\} = 2, \\
 w_3^1(2) &= \max\{0, c_{2\sigma(1)} - \max\{\beta_2, a_{2\sigma(2)} - \max\{0, b_{2\sigma(1)} \\
 &\quad - \beta_2 - w_3^4(2)\}\}\} = X - 2 \dots,
 \end{aligned}$$

thus $w_3^2(q)$ (respectively, $w_3^4(q)$) keeps increasing by one time unit per MPS cycle until it reaches $X + 1$ (resp., X), at which time a steady state has also been reached. In this case it takes X cycles to reach the steady state. $\square$

We now consider cycle S_6 , for which the cycle time expression is derived in Lemma 8. We need the following preliminary result.

Lemma 12. Let $w_1^1, \dots, w_1^n$ and $w_1'^1, \dots, w_1'^n$ denote two feasible vectors of waiting times in S_6 , where $w_1^{i+1} = \max\{0, a_{6\sigma(i+1)} - \max\{\beta_6, b_{6\sigma(i)}, c_{6\sigma(i-1)} - w_1^i\}\}$, and $w_1'^{i+1} = \max\{0, a_{6\sigma(i+1)} - \max\{\beta_6, b_{6\sigma(i)}, c_{6\sigma(i-1)} - w_1'^i\}\}$. If $w_1^i \leq w_1'^i$, then $w_1^{i+1} \leq w_1'^{i+1}$.

Proof. Follows that of Lemma 10. $\square$

Remark 6. It follows from Lemma 12 that, whether n is even or odd, the S_6 cycle consisting of a single MPS is sufficient to define a steady state.

Hall et al. (1997) show that the problem of finding the best part sequence to be processed using robot move cycle S_6 is intractable. They also describe an $O(n^2)$ time algorithm to find the steady state waiting times and the cycle time, for a given part sequence under S_6 .

There are two different ways in which cycle S_6 can be initialized from the state $\hat{E} = (\phi, \phi, \phi, I)$ in $MRC3|k \geq 2, S_6|C_r$:

(a) Initialization **INA6**, starting with part $P_{\sigma(i)}$: pick up $P_{\sigma(i)}: (\epsilon_1)$, move to $M_1: (\delta_1)$, load $P_{\sigma(i)}$ on $M_1: (\epsilon_2)$, wait at $M_1: (w_1^1(0) = a_{\sigma(i)})$, unload $P_{\sigma(i)}$ from $M_1: (\epsilon_3)$, move to $M_2: (\delta_0 + \delta_2)$, load $P_{\sigma(i)}$ on $M_2: (\epsilon_4)$, move to $I: (\delta_0 + \delta_1 + \delta_2 - \gamma)$, pick up $P_{\sigma(i+1)}: (\epsilon_1)$, move to $M_1: (\delta_1)$, load $P_{\sigma(i+1)}$ on $M_1: (\epsilon_2)$, move to $M_2: (\delta_0 + \delta_2)$, if necessary wait at $M_2: (w_2^1(0))$, unload $P_{\sigma(i)}$ from $M_2: (\epsilon_5)$, move to $M_3: (\delta_0 + \delta_3)$, load $P_{\sigma(i)}$ on $M_3: (\epsilon_6)$, move to $M_1: (2\delta_0 + \delta_2 + \delta_3 - \gamma)$, if necessary wait at $M_1: (w_1^{i+1}(0))$, unload $P_{\sigma(i+1)}$ from $M_1: (\epsilon_3)$, move to $M_2: (\delta_0 + \delta_2)$, load $P_{\sigma(i+1)}$ on $M_2: (\epsilon_4)$, move to $I: (\delta_0 + \delta_1 + \delta_2 - \gamma)$, pick up $P_{\sigma(i+2)}: (\epsilon_1)$, move to $M_1: (\delta_1)$, and load $P_{\sigma(i+2)}$ on $M_1: (\epsilon_2)$. At this point, cycle S_6 starts from the state $E = (\Omega, \Omega, \Omega, M_1^-)$.

The waiting time equations corresponding to initialization **INA6** are as follows:

$$\begin{aligned}
 w_1^1(0) &= a_{\sigma(i)}, \\
 w_2^1(0) &= \max\{0, b_{\sigma(i)} - 2\delta_0 - 2\delta_1 - 2\delta_2 - \epsilon_1 - \epsilon_2 + \gamma\}, \\
 w_1^{i+1}(0) &= \max\{0, a_{\sigma(i+1)} - w_2^1(0) - 4\delta_0 - 2\delta_2 - 2\delta_3 \\
 &\quad - \epsilon_5 - \epsilon_6 + \gamma\}.
 \end{aligned}$$

The waiting time equations corresponding to the q th S_6 cycle are as follows, where $q = 1, 2, \dots$, and $w_3^{i+n}(q) = w_3^j(q)$, $j = 1, \dots, n$:

$$\begin{aligned}
 w_1^i(q) &= \max\{0, a_{6\sigma(j)} - \max\{\beta_6, b_{6\sigma(j-1)}, c_{6\sigma(j-2)} \\
 &\quad - w_1^{i-1}(q-1)\}\}, \quad j = i + 2, \\
 w_1^i(q) &= \max\{0, a_{6\sigma(j)} - \max\{\beta_6, b_{6\sigma(j-1)}, c_{6\sigma(j-2)} \\
 &\quad - w_1^{i-1}(q)\}\}, \quad j = i + 3, \dots, i + n + 1.
 \end{aligned}$$

(b) Initialization **INB6**, starting with part $P_{\sigma(i)}$: pick up $P_{\sigma(i)}: (\epsilon_1)$, move to $M_1: (\delta_1)$, load $P_{\sigma(i)}$ on $M_1: (\epsilon_2)$, wait at $M_1: (w_1^1(0) = a_{\sigma(i)})$, unload $P_{\sigma(i)}$ from $M_1: (\epsilon_3)$, move to $M_2: (\delta_0 + \delta_2)$, load $P_{\sigma(i)}$ on $M_2: (\epsilon_4)$, wait at $M_2: (w_2^1(0) = b_{\sigma(i)})$, unload $P_{\sigma(i)}$ from $M_2: (\epsilon_5)$, move to $M_3: (\delta_0 + \delta_3)$, load $P_{\sigma(i)}$ on $M_3: (\epsilon_6)$, move to $I: (2\delta_0 + \delta_1 + \delta_2 + \delta_3 - 2\gamma)$, pick up $P_{\sigma(i+1)}: (\epsilon_1)$, move to $M_1: (\delta_1)$, load $P_{\sigma(i+1)}$ on $M_1: (\epsilon_2)$, wait at $M_1: (w_1^{i+1}(0) = a_{\sigma(i+1)})$, unload $P_{\sigma(i+1)}$ from $M_1: (\epsilon_3)$, move to $M_2: (\delta_0 + \delta_2)$, load $P_{\sigma(i+1)}$ on $M_2: (\epsilon_4)$, move to $I: (\delta_0 + \delta_1 + \delta_2 - \gamma)$, pick up $P_{\sigma(i+2)}: (\epsilon_1)$, move to $M_1: (\delta_1)$, and load $P_{\sigma(i+2)}$ on $M_1: (\epsilon_2)$. At this point, cycle S_6 starts from the state $E = (\Omega, \Omega, \Omega, M_1^-)$.

The waiting time equations corresponding to initialization **INB6** are as follows:

$$\begin{aligned}
w_1^i(0) &= a_{\sigma(i)}, \\
w_2^i(0) &= b_{\sigma(i)}, \\
w_1^{i+1}(0) &= a_{\sigma(i+1)}, \\
w_3^i(0) &= \max\{0, c_{\sigma(i)} - a_{\sigma(i+1)} - 6\delta_0 - 4\delta_1 - 4\delta_2 \\
&\quad - 2\delta_3 - 2\epsilon_1 - 2\epsilon_2 - \epsilon_3 - \epsilon_4 + 4\gamma\}, \\
w_2^{i+1}(1) &= \max\{0, b_{\sigma(i+1)} - w_3^i(0) - 4\delta_0 - 2\delta_1 - 2\delta_2 \\
&\quad - 2\delta_3 - 2\delta_4 - \epsilon_1 - \epsilon_2 - \epsilon_7 \\
&\quad - \epsilon_8 + 3\gamma\}, \\
w_1^{i+2}(1) &= \max\{0, a_{\sigma(i+2)} - w_3^i(0) - w_2^{i+1}(1) - 6\delta_0 \\
&\quad - 2\delta_2 - 4\delta_3 - 2\delta_4 - \epsilon_5 - \epsilon_6 - \epsilon_7 - \epsilon_8 \\
&\quad + 3\gamma\}.
\end{aligned}$$

The waiting time equations corresponding to the q th S_6 cycle are as follows, where $q = 1, 2, \dots$, and $w_3^{j+n}(q) = w_3^j(q)$, $j = 1, \dots, n$:

$$\begin{aligned}
w_1^j(q) &= \max\{0, a_{6\sigma(j)} - \max\{\beta_6, b_{6\sigma(j-1)}, c_{6\sigma(j-2)} \\
&\quad - w_1^{j-1}(q-1)\}\}, \quad j = i+3, \\
w_1^j(q) &= \max\{0, a_{6\sigma(j)} - \max\{\beta_6, b_{6\sigma(j-1)}, c_{6\sigma(j-2)} \\
&\quad - w_1^{j-1}(q)\}\}, \quad j = i+4, \dots, i+n+2.
\end{aligned}$$

The main result for the convergence of cycle S_6 to a steady state now follows.

Theorem 9. *Starting from the state $\hat{E} = (\phi, \phi, \phi, I)$, using part sequence σ , the cell in $MRC3|k \geq 2, S_6|C_r$ will go through at most $\max\{1, \min_{1 \leq i \leq n} \{s_i\}\}$ cycles before reaching a steady state, where $s_i = a_{6\sigma(i)} - \max\{\beta_6, b_{6\sigma(i-1)}\}$, $i = 1, \dots, n$, and this bound is attainable.*

Proof. Similar to that of Theorem 8. $\square$

In this section, we have shown that an MRC3 cell, operating in any one of robot move cycles S_1, S_3, S_4 , or S_5 , reaches a steady state within the production of one MPS, if it starts from an empty cell $\hat{E} = (\phi, \phi, \phi, I)$ where the robot is at I ready to pick up a part. Using robot move cycle S_2 or S_6 starting from $\hat{E}$, the cell converges to a steady state in a number of cycles which is bounded by a function of the cell data. Note that for cycle S_2 (respectively, S_6) the bound derived in Theorem 8 (resp., Theorem 9) is also valid for the cell starting from any other state. Similarly, it is easy to show that a cell operating in any one of robot move cycles S_1, S_3, S_4 , or S_5 reaches a steady state starting from any other state, within the production of at most three MPSs.

5. CONCLUSIONS

In this paper we provide several basic results about the solvability of scheduling problems arising in robotic cells, with the objective of minimizing the cycle time needed for repetitive manufacture of an MPS. In particular, we

describe an algorithm that simultaneously optimizes the robot move cycle and part sequence, for a very general two machine cell. Our proof that simple, repetitive robot move cycles which produce one unit dominate more complicated cycles in single part-type production should greatly simplify the search for good solutions in that context. We also provide several basic results about the solvability of scheduling problems arising in three machine robotic cells. In particular, we show that four out of the six potentially optimal robot move cycles in a general three machine cell allow the efficient identification of an optimal part sequence. Under the other two robot move cycles, however, the recognition version of the part sequencing problem is unary NP-complete (Hall et al. 1997). We identify realistic conditions on the relative lengths of processing times, compared to robot move times, under which the last two cycles are dominated by the other four, and thus the part sequencing problem can still be solved efficiently. The process by which a robotic cell converges to a steady state is also investigated.

Other possible topics for future research include scheduling in robotic cells under uncertainty (for example, with stochastic processing times). While we study an environment requiring the repetitive manufacture of a small variety of similar items, which makes the cycle time criterion a natural one to consider, other criteria such as minimizing the makespan for a fixed set of parts could also be useful. The design of polynomial approximation schemes that can deliver solutions with prespecified accuracy for intractable robotic cell scheduling problems is also important. Other topics of importance include the use of limited buffer storage within the cell as a means of reducing cycle time, and the resulting tradeoffs between throughput rate and inventory holding cost. Also, material handling devices other than robots generate similar problems with related scheduling issues. For example, Matsuo et al. (1991) discuss the similarities between material handling devices in a computer integrated manufacturing environment. Among these devices are cranes, monorails, and AGVs. These devices are frequently used in industry with limited storage buffers. Important insights about various problems arising in environments served by these devices can be gained from a study of robot-served manufacturing cells.

APPENDIX A CYCLE TIME DERIVATIONS

Here we derive the cycle times for cycles $S_1, \dots, S_6$ in $MRC3|k \geq 2, 1 - \text{unit}|C_r$. This environment is more general than that required by Lemma 6, but provides the results needed there for the case $k = 1$ by setting $a_i = a$, $b_i = b$, $c_i = c$, $i = 1, \dots, n$, and $n = 1$. Let $T_{j\sigma(i)\sigma(i+1)}^h$ denote the time between the loading of part $P_{\sigma(i)}$ on machine M_h , and the loading of part $P_{\sigma(i+1)}$ on machine M_h , using robot move cycle S_j , where $h \in \{1, 2, 3\}$ and $j \in \{1, \dots, 6\}$.

$$T_{1\sigma(i)\sigma(i+1)}^1 = \alpha_1 + a_{\sigma(i)} + b_{\sigma(i)} + c_{\sigma(i)},$$

where $\alpha_1 = 4\delta_0 + 2 \sum_{i=1}^4 \delta_i + \sum_{i=1}^8 \epsilon_i - 3\gamma$.

$$T_{2\sigma(i+1)\sigma(i+2)}^2 = \sum_{i=1}^8 \epsilon_i + 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 \\ - 4\gamma + w_1^{i+2} + w_2^{i+1} + w_3^i,$$

where $w_1^{i+2} = \max\{0, a_{2\sigma(i+2)} - \beta_2 - w_2^{i+1}\}$, $w_2^{i+1} = \max\{0, b_{2\sigma(i+1)} - \beta_2 - w_3^i\}$, and $w_3^i = \max\{0, c_{2\sigma(i)} - \beta_2 - w_1^{i+1}\}$. Substituting for w_1^{i+2} and w_2^{i+1} , we obtain $T_{2\sigma(i)}^2 = \sum_{i=1}^n T_{2\sigma(i+1)\sigma(i+2)}^2 = n\alpha_2 + \sum_{i=1}^n \max\{\beta_2, a_{2\sigma(i+2)}, b_{2\sigma(i+1)} - w_3^i\} + \sum_{i=1}^n w_3^i$, where $w_3^i = \max\{0, c_{2\sigma(i)} - \max\{\beta_2, a_{2\sigma(i+1)} - \max\{0, b_{2\sigma(i)} - \beta_2 - w_3^{i-1}\}\}\}$, where we define $b_{2\sigma(n+1)} = b_{2\sigma(1)}$, $a_{2\sigma(n+1)} = a_{2\sigma(1)}$, $a_{2\sigma(n+2)} = a_{2\sigma(2)}$, and where $\alpha_2 = 4\delta_0 + 2\delta_2 + 2\delta_3 - \gamma$, $\beta_2 = \sum_{i=1}^8 \epsilon_i + 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + 2\delta_4 - 3\gamma$, $a_{2\sigma(i)} = a_{\sigma(i)} + \sum_{i=1}^4 \epsilon_i + \sum_{i=7}^8 \epsilon_i + 2\delta_1 + 2\delta_4 - 2\gamma$, $b_{2\sigma(i)} = b_{\sigma(i)} + \sum_{i=3}^6 \epsilon_i$, and $c_{2\sigma(i)} = c_{\sigma(i)} + \sum_{i=1}^2 \epsilon_i + \sum_{i=5}^8 \epsilon_i + 2\delta_1 + 2\delta_4 - 2\gamma$. For the identical part case ($k = 1$), $T_2 = \max\{8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 4\gamma, c + 4\delta_0 + 2 \sum_{i=1}^4 \delta_i + \epsilon_1 + \epsilon_2 + \epsilon_5 + \epsilon_6 + \epsilon_7 + \epsilon_8 - 3\gamma, b + 4\delta_0 + 2\delta_2 + 2\delta_3 + \sum_{i=3}^6 \epsilon_i - \gamma, a + 4\delta_0 + 2 \sum_{i=1}^4 \delta_i + \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4 + \epsilon_7 + \epsilon_8 - 3\gamma, (a + b + c)/2 + 2\delta_0 + \delta_1 + 2\delta_2 + \delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i/2 - 3\gamma/2\}$. A detailed proof of the expression for T_2 is independently provided by Ioachim and Soumis (1995).

$$T_{3\sigma(i)\sigma(i+1)}^3 = a_{\sigma(i+1)} + 6\delta_0 + 2\delta_1 + 2\delta_2 + 4\delta_3 + 2\delta_4 \\ + \sum_{i=1}^8 \epsilon_i - 3\gamma + w_1^{i+1} + w_2^i,$$

where $w_2^{i+1} = \max\{0, b_{\sigma(i+1)} - 2\delta_0 - 2\delta_3 - 2\delta_4 - \epsilon_7 - \epsilon_8 + \gamma - w_3^i\}$, and $w_3^i = \max\{0, c_{\sigma(i)} - a_{\sigma(i+1)} - 4\delta_0 - 2\delta_1 - 2\delta_2 - 2\delta_3 - \epsilon_1 - \epsilon_2 - \epsilon_3 - \epsilon_4 + 2\gamma\}$. This gives

$$T_{3\sigma(i)\sigma(i+1)}^3 = \alpha_3 \\ + \max\{\beta_3 + a_{\sigma(i+1)}, b_{3\sigma(i+1)} + a_{\sigma(i+1)}, c_{3\sigma(i)}\},$$

where $\alpha_3 = 2\delta_0 + 2\delta_3 + \epsilon_5 + \epsilon_6 - \gamma$, $\beta_3 = 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + 2\delta_4 + \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4 + \epsilon_7 + \epsilon_8 - 2\gamma$, $b_{3\sigma(i)} = b_{\sigma(i)} + 2\delta_0 + 2\delta_1 + 2\delta_2 + \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4 - \gamma$, and $c_{3\sigma(i)} = c_{\sigma(i)} + 2\delta_4 + \epsilon_7 + \epsilon_8$.

$$T_{4\sigma(i+1)\sigma(i+2)}^4 = b_{\sigma(i+1)} + 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 \\ + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 5\gamma + w_1^{i+1} + w_2^i,$$

where $w_1^{i+1} = \max\{0, a_{\sigma(i+1)} - w_3^i - 4\delta_0 - 2\delta_2 - 2\delta_3 - 2\delta_4 - \epsilon_7 - \epsilon_8 + 3\gamma\}$, and $w_3^i = \max\{0, c_{\sigma(i)} - 4\delta_0 - 2\delta_1 - 2\delta_2 - 2\delta_3 - \epsilon_1 - \epsilon_2 + 3\gamma\}$. This gives

$$T_{4\sigma(i+1)\sigma(i+2)}^4 = \alpha_4 + b_{\sigma(i+1)} + \max\{\beta_4, c_{4\sigma(i)}, a_{4\sigma(i+1)}\},$$

where $\alpha_4 = 4\delta_0 + 2\delta_2 + 2\delta_3 + \epsilon_3 + \epsilon_4 + \epsilon_5 + \epsilon_6 - 2\gamma$, $\beta_4 = 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + 2\delta_4 + \epsilon_1 + \epsilon_2 + \epsilon_7 + \epsilon_8 - 3\gamma$, $a_{4\sigma(i)} = a_{\sigma(i)} + 2\delta_1 + \epsilon_1 + \epsilon_2$, and $c_{4\sigma(i)} = c_{\sigma(i)} + 2\delta_4 + \epsilon_7 + \epsilon_8$.

$$T_{5\sigma(i+1)\sigma(i+2)}^5 = c_{\sigma(i)} + 6\delta_0 + 2\delta_1 + 4\delta_2 + 2\delta_3 + 2\delta_4 \\ + \sum_{i=1}^8 \epsilon_i - 3\gamma + w_1^{i+1} + w_2^i,$$

where $w_1^{i+1} = \max\{0, a_{\sigma(i+1)} - w_2^i - 4\delta_0 - 2\delta_2 - 2\delta_3 - 2\delta_4 - \epsilon_5 - \epsilon_6 - \epsilon_7 - \epsilon_8 + 2\gamma - c_{\sigma(i)}\}$, and $w_2^i = \max\{0, b_{\sigma(i)} - 2\delta_0 - 2\delta_1 - 2\delta_2 - \epsilon_1 - \epsilon_2 + \gamma\}$. This gives

$$T_{5\sigma(i+1)\sigma(i+2)}^5 = \alpha_5 + \max\{\beta_5 + c_{\sigma(i)}, b_{5\sigma(i)} + c_{\sigma(i)}, a_{5\sigma(i+1)}\},$$

where $\alpha_5 = 2\delta_0 + 2\delta_2 + \epsilon_3 + \epsilon_4 - \gamma$, $\beta_5 = 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + 2\delta_4 + \epsilon_1 + \epsilon_2 + \epsilon_5 + \epsilon_6 + \epsilon_7 + \epsilon_8 - 2\gamma$, $a_{5\sigma(i)} = a_{\sigma(i)} + 2\delta_1 + \epsilon_1 + \epsilon_2$, and $b_{5\sigma(i)} = b_{\sigma(i)} + 2\delta_0 + 2\delta_3 + 2\delta_4 + \epsilon_5 + \epsilon_6 + \epsilon_7 + \epsilon_8 - \gamma$.

$$T_{6\sigma(i+2)\sigma(i+3)}^6 = \sum_{i=1}^8 \epsilon_i + 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 \\ + 2\delta_4 - 4\gamma + w_1^{i+2} + w_2^{i+1} + w_3^i,$$

where $w_1^{i+2} = \max\{0, a_{6\sigma(i+2)} - \beta_6 - w_2^{i+1} - w_3^i\}$, $w_2^{i+1} = \max\{0, b_{6\sigma(i+1)} - \beta_6 - w_3^i\}$, and $w_3^i = \max\{0, c_{6\sigma(i)} - \beta_6 - w_1^{i+1}\}$. Substituting for w_1^{i+2} , w_2^{i+1} and w_3^i , we obtain $T_{6\sigma(i)}^6 = \sum_{i=1}^n T_{6\sigma(i+2)\sigma(i+3)}^6 = n\alpha_6 + \sum_{i=1}^n \max\{\beta_6, c_{6\sigma(i)} - w_1^{i+1}, b_{6\sigma(i+1)}, a_{6\sigma(i+2)}\}$, and $w_1^{i+1} = \max\{0, a_{6\sigma(i+1)} - \max\{\beta_6, b_{6\sigma(i)}, c_{6\sigma(i-1)} - w_1^i\}\}$, where $\alpha_6 = 2\delta_0$, $\beta_6 = 6\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \epsilon_i - 4\gamma$, $a_{6\sigma(i)} = a_{\sigma(i)} + 2\delta_1 + 2\delta_2 + \sum_{i=1}^4 \epsilon_i - \gamma$, $b_{6\sigma(i)} = b_{\sigma(i)} + 2\delta_0 + 2\delta_2 + 2\delta_3 + \sum_{i=3}^6 \epsilon_i - \gamma$, and $c_{6\sigma(i)} = c_{\sigma(i)} + 2\delta_3 + 2\delta_4 + \sum_{i=5}^8 \epsilon_i - \gamma$. For the identical part case considered in Lemma 6, the cycle time, T_6 , can be written as $T_6 = \alpha_6 + \max\{\beta_6, c_6, b_6, a_6\}$. A detailed proof of the expression for T_6 is independently provided by Ioachim and Soumis (1995).

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